l_0-l_1 Hybrid Total Variation Regularization and Its Applications on Hyperspectral Image Mixed Noise Removal and Compressed Sensing

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***Abstract*— The total variation (TV) regularization has been widely used in various applications related to hyperspectral (HS) signal and image processing due to its potential in modeling the underlying smoothness of HS data. However, most existing TV norms usually tend to generate spatial oversmoothing or artifacts. To this end, we propose a novel l_0-l_1 hybrid TV (l_0-l_1HTV) regularization with the applications to HS mixed noise removal and compressed sensing (CS). More specifically, l_0-l_1HTV can be regarded as a globally and locally integrated TV regularizer, where the l_0 gradient constraint is incorporate into the l_1 spatial–spectral TV (l_1-SSTV). l_1-SSTV is capable of exploiting the local structure information across both spatial and spectral domains, while the l_0 gradient can promote a globally spectral–spatial smoothness by directly controlling the number of nonzero gradients of HS images. This efficient combination considers more comprehensive prior knowledge of HS images, yielding sharper edge preservation and resolving the above drawbacks of existing pure TV norms. More significantly, l_0-l_1HTV can be easily injected into HS-related processing models, and an effective algorithm based on the alternating direction method of multipliers (ADMM) is developed to solve the optimization problems. Extensive experiments conducted on several HS data sets substantiate the superiority and effectiveness of the proposed method in comparison with many state-of-the-art methods.**

***Index Terms*— l_0-l_1 hybrid total variation (l_0-l_1HTV), l_0 gradient, l_1 spatial–spectral TV (l_1SSTV), alternating direction method of multipliers (ADMM), compressed sensing (CS), hyperspectral image (HSI) denoising.**

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I. Introduction

HYPERSPECTRAL (HS) imagery acquires hundreds of spectral bands ranging from ultraviolet to infrared wavelength for the same scene on the Earth surface. Compared with other traditional imagery techniques, hyperspectral image (HSI) provides more abundant spatial and spectral information and, thus, has its wide range of applications in agriculture, military, environment monitoring, and food industry [1]–[3]. However, due to various factors of the imaging system and environment, the observed HSIs are unavoidably degraded by noises in real-world scenes. Moreover, in some high-resolution or long-distance transmission HS imaging systems, the increasing amount of 3-D HS data causes the high cost of imaging, storage, or transmission. HSI compressed sensing (CS) is designed to reduce the computation of imaging, and the final recovered HSI images are estimated from a few measurements. As essential preprocessing steps, these HSI restorations have direct influence on the performances of further HSI image processing including classification [4]–[6], unmixing [7], [8], super-resolution [9], [10], feature learning [11], [12], and target detection [13]–[15]. Therefore, it is critical to design effective techniques for HSI restoration.

In general, the optimization problem of HSI restoration methods consists of data fidelity and regularization terms. The former term is derived from a forward model of the object observation. The regularization term can be interpreted as the prior information from underlying properties on recovered HS images. One of the prevalent regularization techniques for image restoration is based on total variation (TV). The TV regularization was introduced to remove noise and preserve image edges for natural image denoising [16]–[18] and CS reconstruction [19]–[21]. Yuan *et al*. [22] explored local spatial prior and developed a spectral–spatial adaptive TV (SSAHTV) model for HSI denoising. However, SSAHTV just reduces the Gaussian noise and loses useful image detail information. Li *et al*. [23] minimized the TV of the abundance fractions subject for HSI CS and unmixing. Eason and Andrews [24] proposed an iterative TV method for HSI CS. However, these TV norms strongly enhance spatial piecewise smoothness, which ignores HSI spectral correlation and leads to spatial oversmoothing.

To overcome the drawbacks of the above TV norms, the spectral correlation of 3-D HSIs is taken into account, and

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many new TV norms are proposed. Ono *et al*. [25] proposed an arranged structure tensor total variation (ASTV) for multichannel image restoration. However, this regularization based on the singular value decomposition of the low-rank (LR) matrix spends much computational time. Chang *et al*. [26] proposed an anisotropic spectral–spatial total variation (ASSTV) regularization to promote the smoothness across both the spectral and spatial differences. Unfortunately, ASSTV causes spectral smoothing due to its strong suppression of direct spectral difference. Aggarwal and Majumdar [27] combined 2-D TV along the spatial dimension and 1-D TV along the spectral dimension and proposed a spatiospectral total variation (SSTV) model. Although SSTV can remove several types of noises, it produces obvious artifacts in the final recovered HSI images. Takeyama *et al*. [28] proposed a hybrid spatiospectral total variation (HSSTV), which is designed to fuse two kinds of local differences with a balancing weight. Specifically, HSSTV1 and HSSTV2 utilize local l_1 spatial norm and l_2 spatial norm, respectively. However, the authors need to analyze and determine an extra parameter and, unfortunately, lose the consideration of HSI global spectral information.

By taking advantage of another HSI prior, i.e., LR, TV regularization is merged with LR models. LR representation reveals that the spectral bands of HSI lie in the same LR space and the HSI data can be represented by a linear combination of finite endmembers. Zhang *et al*. [29] rearranged a 3-D HSI data as a 2-D matrix and proposed an LR matrix approximation (LRMR) method. Subsequently, He *et al*. [30] proposed a band-by-band TV regularized LR matrix factorization (LRTV) method, and Wu *et al*. [31] fused the band-by-band TV regularization into the weighted nuclear norm minimization (WNMM). Golbabaee and Vandergheynst [32] proposed a convex function that penalizes both the trace norm and the band-by-band TV norm of the LR HSI matrix for HSI CS. However, this band-by-band TV regularization handles each HSI band separately and, thus, yields unsatisfying spectral distortions. Wu *et al*. [33] incorporated the structure tensor total variation (STV) into the above WNNM to proposed an STWNNM model. To exploit spatiospectral total variation (SSTV), Wang *et al*. [34] proposed a novel LR constraint and SSTV regularization model, and Wang *et al*. [35] integrated this kind of SSTV into multidirectional LR frameworks. Chen *et al*. [36] proposed a weighted group sparsity regularized $l_{2,1}$ norm to exploit the shared sparse pattern of the difference image for different bands and combined this regularization into the well-known LR Tucker decomposition. By using a 3-D anisotropic total variation (3DATV), He *et al*. [37] combined it with the local LR matrix and proposed an LLRSSTV method for HSI mixed noise removal. Wang *et al*. [38] merged this 3DATV into an LR tensor Tucker decomposition model. Peng *et al*. [39] calculated the sparsity on the subspace of gradient maps and proposed an enhanced 3-D total variation (E-3DTV) regularization for HSI denoising and CS. Wang *et al*. [40] propose a novel tensor-based approach named JTenRe3-DTV for HSI CS, by jointing the tensor Tucker decomposition and 3DTV. Nevertheless, all the above TVs lean upon l_1-norms and penalize large image gradient magnitudes. In real cases, HSI edges can be adversely affected.

To effectively sharpen image edges, Li *et al*. [41] first proposed a new l_0 gradient minimization for natural image smoothing. This minimization globally controlling how many nonzero gradients of results to recover prominent structure have been used in image denoising [41], image deblurring [42], and image inpainting [43]. Xu *et al*. [42] proposed an effective method based on a generalized l_0 sparse expression for motion deblurring. Xue *et al*. [43] extended the above l_0 gradient into a low gradient and converted this regularization into an LR framework. Wang *et al*. [44] developed the l_0 gradient to explore smoothness across both the spatial and spectral domains and combined it with the LR tensor Tucker decomposition model. Xiong *et al*. [45] embedded the l_0 gradient regularization into an LR tensor factorization. However, these LR-based methods are computationally expensive since they require singular value decomposition. Utilizing either spatial smoothness or spatial–spectral smoothness will inevitably cause oversmoothing or artifacts, respectively.

To consider more prior HSI structure information and introduce less computational burden, we combine spatial–spectral TV and l_0 gradient regularization. The aforementioned l_0 gradient has a fatal flaw: a user-give parameter penalizing the degree of smoothness has no physical meaning. Therefore, Ono [46] proposed a novel l_0 gradient projection employing a parameter with a clear meaning: the l_0 gradient value of the output image to control the degree of smoothness. This l_0 gradient projection variant, l_0 HS TV (l_0hTV), is extended for HSI restoration and is effective for extracting image edges information. However, it is difficult for l_0 HS TV to distinguish outliers, such as sparse noise, when these outliers locate on HSI image edges. Based on the abovementioned discussion, we proposed a l_0-l_1 Hybrid Total Variation (l_0-l_1HTV) technique for HSI mixed noise removal and compressed sensing. Compared with existing techniques, the main contributions of this study are summarized as follows.

1) We extend the l_0 gradient minimization into HSI restoration, named l_0hTV, for directly controlling the number of nonzero image gradients and establishing a physical relationship with the l_0 gradient value of the recovered result. l_0hTV characterizes the global spectral–spatial smoothness to complement image details, such as sharp edges and valuable texture.
2) To take full consideration of the local spatial–spectral structure and the global spectral–spatial information of HSI, we design a globally and locally integrated TV regularizer, where l_0hTV is added on SSTV. SSTV captures the local information across spatial and spectral domains to recover the overall structure of HSIs. This TV regularizer compensates mutually to conquer the shortcomings of artifacts or oversmoothing.
3) We integrate SSTV and l_0hTV of HSI into a consolidated framework of l_0-l_1HTV, which can be formulated for two HSI tasks: HSI noise removal and compressed sensing. The Alternating Direction Method of Multipliers (ADMM) is adapted to solve the optimization problems.

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Experimental results with both simulated and real data demonstrate the superiority of l_0-l_1HTV. In particular, the classification accuracy of the results by l_0-l_1HTV is improved significantly.

The remainder of this article is organized as follows. Some notations and preliminaries of tensors are introduced in Section II. Section III presents the study of l_0 gradient minimization and the new l_0-l_1HTV. Then, Sections IV and V provide HSI denoising and compressed sensing models by involving the l_0-l_1HTV regularization, respectively. The experimental results and analysis are reported in Section VI. Finally, the conclusions are drawn in Section VII.

II. Notations and Preliminaries

In this article, scalars, vectors, matrices, and tensors are denoted by lowercase letters, e.g., x, boldface lowercase letters, e.g., $\mathbf{x}$, boldface capital letters, e.g., $\mathbf{X}$, and Euler script letters, e.g., $\mathcal{X}$, respectively. For a three-way HSI tensor $\mathcal{X} \in \mathbb{R}^{h\times v\times z}$, we denote its (h_i, v_i, z_i)-element as $\mathcal{X}_{h_i,v_i,z_i}$ and use notation $\mathcal{X}(i,:,:)$, $\mathcal{X}(:,i,:)$ and $\mathcal{X}(:,:,i)$ to represent the ith horizontal, lateral, and frontal slices, respectively. The l_1-norm is calculated as $||\mathcal{X}||_1 = \sum_{h_i,v_i,z_i} |\mathcal{X}_{h_i,v_i,z_i}|$, and the Frobenius norm is written as $||\mathcal{X}||_F = (\sum_{h_i,v_i,z_i} |\mathcal{X}_{h_i,v_i,z_i}|^2)^{1/2}$. The l_1-norm of a vector denotes $||\mathbf{x}||_1 = \sum_i |\mathbf{x}_i|$, and the l_2-norm of a vector is $||\mathbf{x}||_2 = (\sum_i |\mathbf{x}_i|^2)^{1/2}$. A matrix $\mathbf{X} \in \mathbb{R}^{m\times n}$ can be vectorized as $\mathbf{x} = \text{vec}(\mathbf{X}) \in \mathbb{R}^t$, where $t = m * n$ and vec($\mathbf{X}$) stacks the columns of $\mathbf{X}$.

Definition 1 *(Mixed $l_{1,0}$ Pseudonorm [46]):* Let $\mathbf{y}$ be a vector of $\mathbb{R}^m$, and index sets $\theta_1, \ldots, \theta_i, \ldots, \theta_n (1 \le n \le m)$ satisfy as follows.

1) Each θ_i is a subset of 1,…,m.
2) $\theta_i \cap \theta_l = \emptyset$ for any $i \ne l$.
3) $\cup_{i=1}^n \theta_i = 1, \ldots, m$.

The mixed $l_{1,0}$ pseudonorm of y is defined as follows:

$$||\mathbf{y}||_{1,0}^{\theta} = ||(||\mathbf{y}_{\theta_1}||_1, \ldots, ||\mathbf{y}_{\theta_i}||_1, \ldots, ||\mathbf{y}_{\theta_n}||_1)||_0 \tag{1}$$

where $\mathbf{y}_{\theta_i}$ denotes a subvector of $\mathbf{y}$ with its entries specified by θ_i.

Definition 2 *(Indicator Function $I(\mathbf{y})$ of $l_{1,0}$ Mixed Pseudonorm [46]):*

$$I_{||\mathcal{B}\cdot||_{1,0}^{\theta}}(y) = \begin{cases} 0, & ||\mathcal{B}\mathbf{y}||_{1,0}^{\theta} \le \gamma \\ \infty, & \text{otherwise} \end{cases} \tag{2}$$

where $\mathcal{B}$ is a given operator. In this article, $\mathcal{B}$ is a diagonal tensor with 0 or 1 entries.

Proposition 1 *(Projection Onto $l_{1,0}$ Mixed Pseudonorm Ball With Binary Mask [46]):* Set $\mathbf{y} \in \mathbb{R}^m$ as a known vector and set γ as a nonnegative integer. Let $\mathbf{W}$ be a known diagonal binary matrix, and let $\theta_1, \ldots, \theta_i, \ldots, \theta_n (1 \le n \le m)$ be index sets satisfying the conditions from **definition 1**. Without loss of generality, $\mathbf{Wy} = (\mathbf{y}_{\theta_1}^{\mathrm{T}}, \ldots, \mathbf{y}_{\theta_n}^{\mathrm{T}})^{\mathrm{T}}$ are assumed. $\mathbf{y}_{\theta_1}, \ldots, \mathbf{y}_{\theta_n}$ are sorted in descending order according to $||\mathbf{y}_{\theta_{\vec{1}}}||_2 > ||\mathbf{y}_{\theta_{\vec{2}}}||_2 > \cdots > ||\mathbf{y}_{\theta_{\vec{n}}}||_2$, where $\mathbf{y}_{\theta_{\vec{1}}}, \ldots, \mathbf{y}_{\theta_{\vec{n}}}$ are obtained with the new order, and the original index sets i have $\vec{i}$ that corresponds to it one by one.

For the following problem:

$$\min_{\mathbf{y}^*} ||\mathbf{y}^* - \mathbf{y}||^2 \quad \text{s.t. } ||\mathbf{W}\mathbf{y}^*||_{1,0}^{\theta} \le \gamma \tag{3}$$

one of the optimal solutions is given by

$$\mathbf{y}^* = \begin{cases} \mathbf{y}, & ||\mathbf{W}\mathbf{y}||_{1,0}^{\theta} \le \gamma \\ \left(\tilde{\mathbf{y}}_{\theta_{\vec{1}}}^{\mathrm{T}}, \ldots, \tilde{\mathbf{y}}_{\theta_{\vec{n}}}^{\mathrm{T}}\right)^{\mathrm{T}} + (\mathbf{I} - \mathbf{W})\mathbf{y}, & ||\mathbf{W}\mathbf{y}||_{1,0}^{\theta} > \gamma \end{cases} \tag{4}$$

where

$$\tilde{\mathbf{y}}_{\theta_{\vec{i}}}^{\mathrm{T}} = \begin{cases} \mathbf{y}_{\theta_i}^{\mathrm{T}}, & \text{if } i \in \{1, \ldots, \gamma\} \\ 0, & \text{if } i \in \{\gamma + 1, \ldots, n\}\ . \end{cases} \tag{5}$$

Definition 3 *(Mode-k_1, k_2 Unfolding of a Tensor [47]):* For a m-way tensor $\mathcal{X} \in \mathbb{R}^{n_1\times n_2\times,\ldots,\times n_m}$, this operator is denoted by $\mathcal{X}_{(k_1,k_2)} \in \mathbb{R}^{n_{k_1}\times k_2\times\prod_{i\ne k_1,k_2} n_i}$, whose frontal slices are the lexicographic ordering of the mode-k_1, k_2 slice of $\mathcal{X}$.

III. l_0-l_1 Hybrid Total Variation

A. l_0 Hyperspectral TV

Xu *et al.* [42] first proposed the l_0 gradient model to preserve natural image edges. For a natural 2-D image $\mathbf{X} \in \mathbb{R}^{m\times n}$, the l_0 Gradient is defined as follows [42]:

$$\begin{aligned} l_0\text{TV}(\mathbf{X}) &= \sum^{m*n} C(||\text{vec}(D_h\mathbf{X})||_1 + ||\text{vec}(D_v\mathbf{X})||_1) \\ &= \sum_i^m \sum_j^n C(|\mathbf{X}_{i+1,j} - \mathbf{X}_{i,j}| + |\mathbf{X}_{i,j+1} - \mathbf{X}_{i,j}|) \end{aligned} \tag{6}$$

where D_h and D_v are linear difference operators along image horizontal and vertical directions. $C(\mathbf{X})$ is a binary function counting the number of nonzero image gradients and $C(\mathbf{X}_{i,j}) = 1$, if $\mathbf{X}_{i,j} \ne 0$; $C(\mathbf{X}_{i,j}) = 0$, otherwise.

For an HSI $\mathcal{X} \in \mathbb{R}^{h\times v\times z}$, $\sum_{k=1}^z l_0\text{TV}(\mathcal{X}^k)$ can be a band-by-band model by the simplest way to extend. Although the spatial smoothness of each HSI band is utilized separately, HSI spectral correlation is neglected. Simultaneously, different HSI bands are acquired from the same scene, and the spatial information of all spectral bands should be fused and formulated. l_0hTV is defined as follows:

$$\begin{aligned} l_0\text{hTV}(\mathcal{X}) = &\sum_i^h \sum_j^v C \\ &\left(\sum_k^z (|\mathcal{X}_{i+1,j,k} - \mathcal{X}_{i,j,k}| + |\mathcal{X}_{i,j+1,k} - \mathcal{X}_{i,j,k}|)\right) \end{aligned} \tag{7}$$

where boundary values of gradients are written as $\mathcal{X}_{i,j+1,k} - \mathcal{X}_{i,j,k} = 0$, if $j = v$, and $\mathcal{X}_{i+1,j,k} - \mathcal{X}_{i,j,k} = 0$, if $i = h$. l_0hTV($\mathcal{X}$), actually counting the nonzero gradients in the spatial dimension with the assistance of all spectral bands, greatly benefits for capturing the spatial edges of the HSI.

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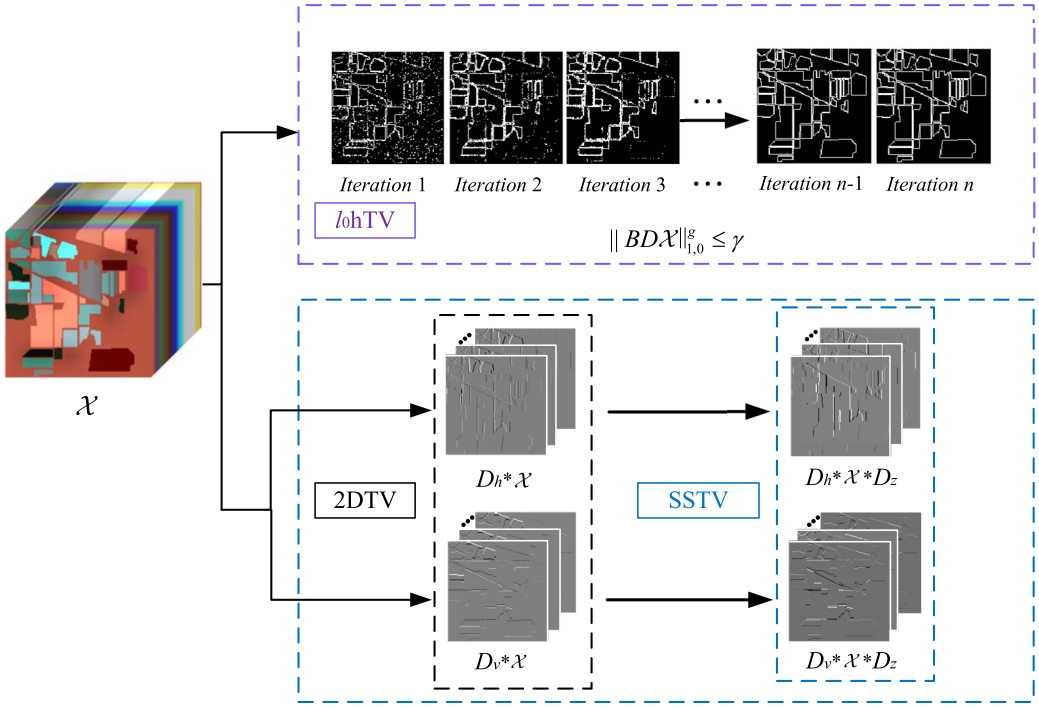


Fig. 1. Illustration of the proposed l_0-l_1 hybrid total variation regularization.

For the mixed $l_{1,0}$ pseudonorm defined by Definition 1, another formulation is adopted to express l_0hTV as follows:

$$l_0\text{hTV}(\mathcal{X}) = ||BD\mathcal{X}||^{\theta}_{1,0} \tag{8}$$

where operator D is an operator to fuse the differences along the horizontal and vertical directions into one tensor. Operator B is an operator that forces boundary values of gradients to be zero when $i = h$ and $j = v$. A simple example is given as follows. $D_h\mathcal{X}$ and $D_v\mathcal{X}$ denote the difference along image horizontal and vertical directions, respectively. $D_h\mathcal{X}$ and $D_v\mathcal{X}$ are stored into one tensor $D\mathcal{X}$, i.e., $D\mathcal{X}(:,:,1,:) = D_h\mathcal{X}$ and $D\mathcal{X}(:,:,2,:) = D_v\mathcal{X}$. The boundary values of $BD\mathcal{X}(:,:,1,1)$, $BD\mathcal{X}(:,:,1,2)$, and $BD\mathcal{X}(:,:,2,1)$, $BDX(:,:,2,2)$ are set as 0 when $i = h = 3$ and $j = v = 3$, respectively.

Example 1: A tensor $\mathcal{X} \in \mathbb{R}^{3\times3\times2}$ and its $BD\mathcal{X} \in \mathbb{R}^{3\times3\times2\times2}$ can be expressed as follows:

$$\mathcal{X}(:,:,1) = \begin{bmatrix} \mathcal{X}_{111} & \mathcal{X}_{121} & \mathcal{X}_{131} \\ \mathcal{X}_{211} & \mathcal{X}_{221} & \mathcal{X}_{231} \\ \mathcal{X}_{311} & \mathcal{X}_{321} & \mathcal{X}_{331} \end{bmatrix} \tag{9}$$

$$\mathcal{X}(:,:,2) = \begin{bmatrix} \mathcal{X}_{112} & \mathcal{X}_{122} & \mathcal{X}_{132} \\ \mathcal{X}_{212} & \mathcal{X}_{222} & \mathcal{X}_{232} \\ \mathcal{X}_{312} & \mathcal{X}_{322} & \mathcal{X}_{332} \end{bmatrix} \tag{10}$$

$$(BD\mathcal{X})(:,:,1,1) = \begin{bmatrix} \mathcal{X}_{121}-\mathcal{X}_{111} & \mathcal{X}_{131}-\mathcal{X}_{121} & 0 \\ \mathcal{X}_{221}-\mathcal{X}_{211} & \mathcal{X}_{231}-\mathcal{X}_{221} & 0 \\ \mathcal{X}_{321}-\mathcal{X}_{311} & \mathcal{X}_{331}-\mathcal{X}_{321} & 0 \end{bmatrix} \tag{11}$$

$$(BD\mathcal{X})(:,:,1,2) = \begin{bmatrix} \mathcal{X}_{122}-\mathcal{X}_{112} & \mathcal{X}_{132}-\mathcal{X}_{122} & 0 \\ \mathcal{X}_{222}-\mathcal{X}_{212} & \mathcal{X}_{232}-\mathcal{X}_{222} & 0 \\ \mathcal{X}_{322}-\mathcal{X}_{312} & \mathcal{X}_{332}-\mathcal{X}_{322} & 0 \end{bmatrix} \tag{12}$$

$$(BD\mathcal{X})(:,:,2,1) = \begin{bmatrix} \mathcal{X}_{211}-\mathcal{X}_{111} & \mathcal{X}_{221}-\mathcal{X}_{121} & \mathcal{X}_{231}-\mathcal{X}_{131} \\ \mathcal{X}_{311}-\mathcal{X}_{211} & \mathcal{X}_{321}-\mathcal{X}_{221} & \mathcal{X}_{331}-\mathcal{X}_{231} \\ 0 & 0 & 0 \end{bmatrix} \tag{13}$$

$$(BD\mathcal{X})(:,:,2,2) = \begin{bmatrix} \mathcal{X}_{212}-\mathcal{X}_{112} & \mathcal{X}_{222}-\mathcal{X}_{122} & \mathcal{X}_{232}-\mathcal{X}_{132} \\ \mathcal{X}_{312}-\mathcal{X}_{212} & \mathcal{X}_{322}-\mathcal{X}_{222} & \mathcal{X}_{332}-\mathcal{X}_{232} \\ 0 & 0 & 0 \end{bmatrix}. \tag{14}$$

With Definition 1, here, $n = 9$, and each $\mathbf{y}_{\theta_j}$ is determined by mode-1,2 unfolding $(BD\mathcal{X})_{(1,2)} \in \mathbb{R}^{9\times2\times2}$, for example, $\mathbf{y}_{\theta_1} = [\mathcal{X}_{121}-\mathcal{X}_{111}, \mathcal{X}_{122}-\mathcal{X}_{112}, \mathcal{X}_{211}-\mathcal{X}_{111}, \mathcal{X}_{212}-\mathcal{X}_{112}]^{\mathrm{T}}$. It is easily proven that l_0hTV$(\mathcal{X})$ and $||BD\mathcal{X}||^{\theta}_{1,0}$ are mathematically equivalent.

B. l_0-l_1 Hybrid Total Variation Regularization

Fig. 1 represents the visual illustration of the l_0-l_1HTV regularization. As shown at the top of Fig. 1, the global spatial smoothness of all spectral bands is restrained by $l_0\text{hTV}(\mathcal{X}) = ||BD\mathcal{X}||^{\theta}_{1,0}$, where γ denotes the l_0 gradient value of the output image. As the number of iterations increases, the image edges of HSI gradually become clearer. Meanwhile, if l_0-norm is excessively penalized, the final results will appear oversmoothing phenomenon. At the bottom of Fig. 1, the SSTV handling local spatial–spectral differences combine 2-D spatial TV (defined by 2DTV$= ||D_h\mathcal{X}||_1 + ||D_v\mathcal{X}||_1$) with 1-D spectral TV. Although most of structure information is restored, SSTV unavoidably caused artifacts. For more comprehensively exploiting the global spatial and the local spatial–spectral information, l_0hTV$(\mathcal{X})$ and SSTV are incorporated as a globally and locally integrated regularizer, and with **Definition 2**, the proposed l_0-l_1 Hybrid Total Variation (l_0-l_1HTV) is defined as follows:

$$\begin{aligned} l_0\text{-}l_1\text{HTV}(\mathcal{X}) &= l_0\text{hTV}(\mathcal{X}) + \text{SSTV}(\mathcal{X}) \\ &= I_{||B\cdot||^{\theta}_{1,0}}(BD\mathcal{X}) + (||D_h\mathcal{X}D_z||_1 + ||D_v\mathcal{X}D_z||_1) \end{aligned} \tag{15}$$

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where D_h and D_v are two operators to calculate differences along the horizontal and vertical directions. D_z is a 1-D finite differencing operator to extract the spectral signature of each pixel. In practical applications, it is hard to find a proper parameter λ for minimizing $\lambda l_0\text{hTV}(\mathcal{X})$ since this user-given parameter λ has no relationship with the degree of the restored image $\mathcal{X}$. Therefore, $||BD\mathcal{X}||_{1,0}^{\theta} \leq \gamma$ is used for HSI restoration, and γ directly represents the l_0 gradient value of the output image.

IV. l_0-l_1 Hybrid Total Variation for HSI Denoising

A clear HSI cube $\mathcal{X} \in \mathbb{R}^{h\times v\times z}$ is corrupted by mixed noise, including the Gaussian noise, salt and pepper noise, deadlines, and strips. The Gaussian noise can be denoted by $\mathcal{N}$, and the others are unified as sparse noise $\mathcal{S}$. Thus, an observed 3D HSI $\mathcal{T} \in \mathbb{R}^{h\times v\times z}$ is represented as follows:

$$\mathcal{T} = \mathcal{X} + \mathcal{S} + \mathcal{N}. \tag{16}$$

With (15) in mind, the proposed l_0-l_1HTV for removing HSI mixed noise is formulated as

$$\min_{\mathcal{X},\mathcal{S}} \|\mathcal{T}-\mathcal{X}-\mathcal{S}\|_F^2 + \lambda\|\mathcal{S}\|_1 + \lambda_{tv} \times (||D_h\mathcal{X}D_z||_1 + ||D_v\mathcal{X}D_z||_1) + I_{||B\cdot||_{1,0}^{\theta}}(BD\mathcal{X}) \tag{17}$$

where λ and λ_{tv} are two positive constants to balance the contributions of each term. The first term $\|\mathcal{T}-\mathcal{X}-\mathcal{S}\|_F^2$ is a data fidelity term. $\|\mathcal{S}\|_1$ is the l_1-norm of the sparse noise $\mathcal{S}$ to remove salt and pepper noise, deadlines, and strips. $||D_h\mathcal{X}D_z||_1 + ||D_v\mathcal{X}D_z||_1$ is used to characterize the local smoothness in the spatial and spectral directions. $I_{||B\cdot||_{1,0}^{\theta}}(BD\mathcal{X})$ is the mixed $l_{1,0}$ pseudonorm to enhance image edges and remove the Gaussian noise.

To solve the above HSI denoising problem, three auxiliary variable $\mathcal{V}$, $\mathcal{P}$, and $\mathcal{Q}$ are introduced, and (17) is reformulated as follows:

$$\begin{aligned}\min_{\mathcal{X},\mathcal{S},\mathcal{V},\mathcal{P},\mathcal{Q}} &\|\mathcal{T}-\mathcal{X}-\mathcal{S}\|_F^2 + \lambda\|\mathcal{S}\|_1 + \lambda_{tv}(||\mathcal{P}||_1 + ||\mathcal{Q}||_1) \\ &+ I_{||B\cdot||_{1,0}^{\theta}}(\mathcal{V}) \\ \text{s.t. } &\mathcal{P} = D_h\mathcal{X}D_z, \mathcal{Q} = D_v\mathcal{X}D_z, \mathcal{V} = D\mathcal{X}.\end{aligned} \tag{18}$$

The ADMM method [48], [49] is applied for the aforementioned problem, and the unconstrained function of (18) is rewritten as:

$$\begin{aligned}\min_{\mathcal{X},\mathcal{S},\mathcal{V},\mathcal{P},\mathcal{Q}} &\|\mathcal{T}-\mathcal{X}-\mathcal{S}\|_F^2 + \lambda\|\mathcal{S}\|_1 + \lambda_{tv}(||\mathcal{P}||_1 + ||\mathcal{Q}||_1) \\ &+ I_{||B\cdot||_{1,0}^{\theta}}(\mathcal{V}) + \beta||\mathcal{P} - D_h\mathcal{X}D_z - \mathcal{Y}_1||_F^2 \\ &+ \beta||\mathcal{Q} - D_v\mathcal{X}D_z - \mathcal{Y}_2||_F^2 + \beta||\mathcal{V} - D\mathcal{X} - \mathcal{Y}_3||_F^2.\end{aligned} \tag{19}$$

Equation (19) is divided into five subproblems:

$$\text{1) } \min_{\mathcal{X}} \|\mathcal{T}-\mathcal{X}-\mathcal{S}\|_F^2 + \beta||\mathcal{P} - D_h\mathcal{X}D_z - \mathcal{Y}_1||_F^2 + \beta||\mathcal{Q} - D_v\mathcal{X}D_z - \mathcal{Y}_2||_F^2 + \beta||\mathcal{V} - D\mathcal{X} - \mathcal{Y}_3||_F^2 \tag{20}$$

$$\text{2) } \min_{\mathcal{S},} \lambda\|\mathcal{S}\|_1 + \|\mathcal{T}-\mathcal{X}-\mathcal{S}\|_F^2 \tag{21}$$

$$\text{3) } \min_{\mathcal{P}} \lambda_{tv}||\mathcal{P}||_1 + \beta||\mathcal{P} - D_h\mathcal{X}D_z - \mathcal{Y}_1||_F^2 \tag{22}$$

$$\text{4) } \min_{\mathcal{Q}} \lambda_{tv}||\mathcal{Q}||_1 + \beta||\mathcal{Q} - D_v\mathcal{X}D_z - \mathcal{Y}_2||_F^2 \tag{23}$$

$$\text{5) } \min_{\mathcal{V}} I_{||B\cdot||_{1,0}^{\theta}}(\mathcal{V}) + \beta||\mathcal{V} - D\mathcal{X} - \mathcal{Y}_3||_F^2. \tag{24}$$

Algorithm 1 l_0-l_1HTV for HSI Mixed Noise Removal

Input: $\mathcal{T}, \lambda, \lambda_{tv}, \gamma$

1: **Initializtion**: $\mathcal{X} = \mathcal{S} = 0$, $\mathcal{V} = \mathcal{P} = \mathcal{Q} = \mathcal{Y}_1 = \mathcal{Y}_2 = \mathcal{Y}_3 = 0$, $i = 0, \epsilon = 10^{-4} * hv$
2: **while** $0 \leq i \leq maxiter$ or $|\ ||BD\mathcal{X}||_{1,0}^{\theta} - \gamma\ | > \epsilon$
3: $i = i + 1$;
4: update $\mathbf{x}$ by solving (26):
$(1 + \beta D^*D + \beta D_1^*D_1 + \beta D_2^*D_2)\mathbf{x} = \mathbf{t} - \mathbf{s}$
5: $+\beta(\mathbf{p} - \mathbf{y}_1 + \mathbf{q} - \mathbf{y}_2 + \mathbf{v} - \mathbf{y}_3)$
6: update $\mathcal{X}$: $\mathcal{X}$=rearrange($\mathbf{x}$)
7: update $\mathcal{S}$, $\mathcal{P}$ and $\mathcal{Q}$ by solving (30), (31) and (32):
8: $\mathcal{S} = S_{\lambda/2}[\mathcal{T} - \mathcal{X}]$
9: $\mathcal{P} = S_{\lambda_{tv}/2}[D_h\mathcal{X}D_z + \mathcal{Y}_1]$
10: $\mathcal{Q} = S_{\lambda_{tv}/2}[D_v\mathcal{X}D_z + \mathcal{Y}_2]$
11: update $\mathcal{V}$ by solving (33):
12: $\mathcal{V}' = B(D\mathcal{X} + \mathcal{Y}_3)$,
13: Sort $\mathcal{V}'_{\theta_1}, \ldots, \mathcal{V}'_{\theta_n}$ in descending order in terms of l_2 norms,
14: Set $\mathcal{V}'_{\theta_{\gamma+1}} = 0, \ldots, \mathcal{V}'_{\theta_n} = 0$ and then obtain a updated $\mathcal{V}'$,
15: $\mathcal{V} = \mathcal{V}' + (I - B)(D\mathcal{X} + \mathcal{Y}_3)$
16: update the multipliers
$\mathcal{Y}_1 = \mathcal{Y}_1 + (D_h\mathcal{X}D_z - \mathcal{P})$
$\mathcal{Y}_2 = \mathcal{Y}_2 + (D_h\mathcal{X}D_z - \mathcal{Q})$
$\mathcal{Y}_3 = \mathcal{Y}_3 + (D\mathcal{X} - \mathcal{V})$
17: **end while**

Output: $\mathcal{X}$.

The $\mathcal{X}$-related subproblem is rewritten into its vector

$$\begin{aligned}\mathbf{x} &= \arg\min_{\mathbf{x}} f(\mathbf{x}) \\ &= \arg\min_{\mathbf{x}} \|\mathbf{t} - \mathbf{x} - \mathbf{s}\|_F^2 + \beta||\mathbf{p} - D_1\mathbf{x} - \mathbf{y}_1||_F^2 \\ &\quad + \beta||\mathbf{q} - D_2\mathbf{x} - \mathbf{y}_2||_F^2 + \beta||\mathbf{v} - D\mathbf{x} - \mathbf{y}_3||_F^2\end{aligned} \tag{25}$$

where $D_1 = D_h \otimes D_z$, $D_2 = D_v \otimes D_z$, and $\otimes$ denotes the Kronecker product. Set the partial derivative of $f(\mathbf{x})$ as zero, and yield the following linear equation:

$$\begin{aligned}&(1 + \beta D^*D + \beta D_1^*D_1 + \beta D_2^*D_2)\mathbf{x} \\ &\qquad = \mathbf{t} - \mathbf{s} + \beta(\mathbf{p} - \mathbf{y}_1 + \mathbf{q} - \mathbf{y}_2 + \mathbf{v} - \mathbf{y}_3).\end{aligned} \tag{26}$$

Equation (26) can be solved by the preconditioned conjugate gradient (PCG) [50] or the algorithm for sparse linear equations and least squares (LSQR) [51], [52].

Equations (21), (22), and (23) about $\mathcal{S}$, $\mathcal{P}$, and $\mathcal{Q}$ are the same form and, respectively, rewritten as

$$\min_{\mathcal{S}} \lambda\|\mathcal{S}\|_1 + \|\mathcal{S} - (\mathcal{T} - \mathcal{X})\|_F^2 \tag{27}$$

$$\min_{\mathcal{P}} \lambda_{tv}||\mathcal{P}||_1 + \beta||\mathcal{P} - (D_h\mathcal{X}D_z + \mathcal{Y}_1)||_F^2 \tag{28}$$

$$\min_{\mathcal{Q}} \lambda_{tv}||\mathcal{Q}||_1 + \beta||\mathcal{Q} - (D_v\mathcal{X}D_z + \mathcal{Y}_2)||_F^2 \tag{29}$$

where the sign operator $sgn(x)$ is defined as follows:

$$\text{sgn}(x) = \begin{cases} 1, & x > 0 \\ 0, & x = 0 \\ -1, & x < 0. \end{cases}$$

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By using the operator $S_\varepsilon[x] = sgn(x).*max(|x|-\varepsilon, 0)$, the solutions are obtained as

$$\mathcal{S} = S_{\lambda/2}[\mathcal{T} - \mathcal{X}] \tag{30}$$

$$\mathcal{P} = S_{\lambda_{tv}/2}[D_h \mathcal{X} D_z + \mathcal{Y}_1] \tag{31}$$

$$\mathcal{Q} = S_{\lambda_{tv}/2}[D_v \mathcal{X} D_z + \mathcal{Y}_2]. \tag{32}$$

The subproblem of $\mathcal{V}$ is equivalent to the following constrained minimization problem:

$$\min_{\mathcal{V}} ||\mathcal{V} - D\mathcal{X} - \mathcal{Y}_3||_F^2 \quad \text{s.t. } ||B\mathcal{V}||_{1,0}^\theta \le \gamma. \tag{33}$$

The parameter β is omitted because it has no impact on the optimization. Proposition 1 is adopted to solve (33). First, $\mathcal{V}'$ is initialized to $\mathcal{V}' = B(D\mathcal{X} + \mathcal{Y}_3)$, and at the ith iteration, $\mathcal{V}'^{(i)}$ is sorted in descending order. $\mathcal{V}'_{\theta_{\gamma+1}} = 0, \ldots, \mathcal{V}'_{\theta_n}$ are set as zero vectors, which effectively removes most of noise while preserving sharper image edges. Simultaneously, artifacts of the denoised result are avoided since the information from all spectral bands is used to promote spatial smoothness. The detail of the l_0-l_1HTV method for HSI mixed noise removal is summarized in Algorithm 1.

V. l_0-l_1 Hybrid Total Variation for HSI Compressed Sensing

HSI compressed sensing aims at reconstructing a 3-D HSI data $\mathcal{X}$ from compressive measurements $\mathbf{y} \in \mathbb{R}^m$. In real acquisition, disturbed by the Gaussian noise $\mathbf{n}$, the compressive measurements $\mathbf{y}$ can be obtained by

$$\mathbf{y} = \Psi(\mathcal{X}) + \mathbf{n}. \tag{34}$$

Ψ is instantiated as $\Psi = \mathbf{D} \cdot \mathbf{H} \cdot \mathbf{P}$, where $\mathbf{D}$ is a random down sampling operator, $\mathbf{H}$ is a random permutation matrix, $\mathbf{P}$ is the Walsh–Hadamard transform, and the mapping of Ψ is $\mathbb{R}^{h\times v\times z} \to \mathbb{R}^m$. This compressed operator has been successfully applied for various CS cases [32], [53], [54]. However, it is an ill-posed inverse problem to recover $\mathcal{X}$ from (34). As the prior information is efficient to find the optimal solution, l_0-l_1HTV regularization is adopted, and the proposed HSI CS model is formulated as

$$\min_{\mathcal{X}} \|\mathbf{y} - \Psi(\mathcal{X})\|_F^2 + \lambda_{tv}(||D_h\mathcal{X}D_z||_1 + ||D_v\mathcal{X}D_z||_1) + I_{||B\cdot||_{1,0}^\theta}(BD\mathcal{X}). \tag{35}$$

By introducing three auxiliary variables $\mathcal{P}$, $\mathcal{Q}$ and $\mathcal{V}$, we rewrite the (35) as follows:

$$\min_{\mathcal{X},\mathcal{V},\mathcal{P},\mathcal{Q}} \|\mathbf{y} - \Psi(\mathcal{X})\|_F^2 + \lambda_{tv}(||\mathcal{P}||_1 + ||\mathcal{Q}||_1) + I_{||B\cdot||_{1,0}^\theta}(\mathcal{V}) \quad \text{s.t. } \mathcal{P} = D_h\mathcal{X}D_z, \mathcal{Q} = D_v\mathcal{X}D_z, \mathcal{V} = D\mathcal{X}. \tag{36}$$

Based on the ADMM method [48], the unconstrained function of (36) is given as follows:

$$\min_{\mathcal{X},\mathcal{V},\mathcal{P},\mathcal{Q}} \|\mathbf{y} - \Psi(\mathcal{X})\|_F^2 + \lambda_{tv}(||\mathcal{P}||_1 + ||\mathcal{Q}||_1) + I_{||B\cdot||_{1,0}^\theta}(\mathcal{V}) + \beta||\mathcal{P} - D_h\mathcal{X}D_z - \mathcal{Y}_1||_F^2 + \beta||\mathcal{Q} - D_v\mathcal{X}D_z - \mathcal{Y}_2||_F^2 + \beta||\mathcal{V} - D\mathcal{X} - \mathcal{Y}_3||_F^2. \tag{37}$$

Equation (37) is divided into four subproblems:

$$1)\ \min_{\mathcal{X}} \|\mathbf{y} - \Psi(\mathcal{X})\|_F^2 + \beta||\mathcal{P} - D_h\mathcal{X}D_z - \mathcal{Y}_1||_F^2 + \beta||\mathcal{Q} - D_v\mathcal{X}D_z - \mathcal{Y}_2||_F^2 + \beta||\mathcal{V} - D\mathcal{X} - \mathcal{Y}_3||_F^2 \tag{38}$$

$$2)\ \min_{\mathcal{P}} \lambda_{tv}||\mathcal{P}||_1 + \beta||\mathcal{P} - D_h\mathcal{X}D_z - \mathcal{Y}_1||_F^2 \tag{39}$$

$$3)\ \min_{\mathcal{Q}} \lambda_{tv}||\mathcal{Q}||_1 + \beta||\mathcal{Q} - D_v\mathcal{X}D_z - \mathcal{Y}_2||_F^2. \tag{40}$$

$$4)\ \min_{\mathcal{V}} I_{||B\cdot||_{1,0}^\theta}(\mathcal{V}) + \beta||\mathcal{V} - D\mathcal{X} - \mathcal{Y}_3||_F^2. \tag{41}$$

Algorithm 2 l_0-l_1HTV for HSI Compressed Sensing

Input: $\mathbf{y}, \lambda_{tv}, \gamma$

Initializtion: $\mathcal{X} = 0$, $\mathcal{V} = \mathcal{P} = \mathcal{Q} = \mathcal{Y}_1 = \mathcal{Y}_2 = \mathcal{Y}_3 = 0$, $i = 0, \epsilon = 10^{-4} * hv$

while $0 \le i \le maxiter$ or $|\ ||BD\mathcal{X}||_{1,0}^\theta - \gamma\ | > \epsilon$

  $i = i + 1$;

  update $\mathbf{x}$ by solving (43):

  $(1 + \beta D^*D + \beta D_1^*D_1 + \beta D_2^*D_2)\mathbf{x} = \mathbf{y} - \Psi(\mathbf{x})$

  $+\beta(\mathbf{p} - \mathbf{y}_1 + \mathbf{q} - \mathbf{y}_2 + \mathbf{v} - \mathbf{y}_3)$

  update $\mathcal{X}$: $\mathcal{X}$=rearrange($\mathbf{x}$)

  update $\mathcal{P}$ and $\mathcal{Q}$ by solving (44) and (45):

  $\mathcal{P} = S_{\lambda_{tv}/2}[D_h\mathcal{X}D_z + \mathcal{Y}_1]$

  $\mathcal{Q} = S_{\lambda_{tv}/2}[D_v\mathcal{X}D_z + \mathcal{Y}_2]$

  update $\mathcal{V}$ by solving (33):

  $\mathcal{V}' = B(D\mathcal{X} + \mathcal{Y}_3)$,

  Sort $\mathcal{V}'_{\theta_1}, \ldots, \mathcal{V}'_{\theta_n}$ in descending order in terms of l_2 norms,

  Set $\mathcal{V}'_{\theta_{\gamma+1}} = 0, \ldots, \mathcal{V}'_{\theta_n} = 0$ and then obtain a updated $\mathcal{V}'$,

  $\mathcal{V} = \mathcal{V}' + (I - B)(D\mathcal{X} + \mathcal{Y}_3)$

  update the multipliers

  $\mathcal{Y}_1 = \mathcal{Y}_1 + (D_h\mathcal{X}D_z - \mathcal{P})$

  $\mathcal{Y}_2 = \mathcal{Y}_2 + (D_h\mathcal{X}D_z - \mathcal{Q})$

  $\mathcal{Y}_3 = \mathcal{Y}_3 + (D\mathcal{X} - \mathcal{V})$

end while

Output: $\mathcal{X}$.

The $\mathcal{X}$-related subproblem is rewritten into its vector

$$\begin{aligned}\mathbf{x} &= \arg\min_{\mathbf{x}} f(\mathbf{x}) \\ &= \arg\min_{\mathbf{x}} \|\mathbf{y} - \Psi(\mathbf{x})\|_F^2 + \beta||\mathbf{p} - D_1\mathbf{x} - \mathbf{y}_1||_F^2 \\ &\quad + \beta||\mathbf{q} - D_2\mathbf{x} - \mathbf{y}_2||_F^2 + \beta||\mathbf{v} - D\mathbf{x} - \mathbf{y}_3||_F^2.\end{aligned} \tag{42}$$

The partial derivative of $f(\mathbf{x})$ is set as zero, and optimizing $\mathbf{x}$ is equivalent to solve the following linear equation:

$$(1 + \beta D^*D + \beta D_1^*D_1 + \beta D_2^*D_2)\mathbf{x} = \mathbf{y} - \Psi(\mathbf{x}) + \beta(\mathbf{p} - \mathbf{y}_1 + \mathbf{q} - \mathbf{y}_2 + \mathbf{v} - \mathbf{y}_3). \tag{43}$$

Equation (43) is also solved by the PCG [50] or the algorithm for sparse linear equations and least squares (LSQR) [52]. Based on the operator $S_\varepsilon[x] = \text{sgn}(x).*\max(|x| - \varepsilon, 0)$, the solutions of (39) and (40) are calculated as

$$\mathcal{P} = S_{\lambda_{tv}/2}[D_h\mathcal{X}D_z + \mathcal{Y}_1] \tag{44}$$

$$\mathcal{Q} = S_{\lambda_{tv}/2}[D_v\mathcal{X}D_z + \mathcal{Y}_2]. \tag{45}$$

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TABLE I
ADDED MIXED NOISES IN NINE CASES

Noise Case	Mixed Noise
Case 1	Gaussian (SNR = $10 \sim 20$ dB)
Case 2	Gaussian (SNR = $10 \sim 20$ dB)+salt and pepper(10%) in all bands
Case 3	Gaussian (SNR = $10 \sim 20$ dB)+salt and pepper(20%) in all bands + four vertical and horizontal deadlines in each of four random bands
Case 4	Gaussian (mean SNR = 12.12 dB, and noise variance is calculated by the Eq.(50))
Case 5	Gaussian (mean SNR = 12.12 dB, and noise variance is calculated by the Eq.(50)) +salt and pepper(10%) in all bands
Case 6	Gaussian (mean SNR = 12.12 dB, and noise variance is calculated by the Eq.(50))+salt and pepper(20%) in all bands + eight vertical and horizontal deadlines in each of eight random bands
Case 7	Gaussian (zero-mean, and noise variance is different for different bands and is selected from 0.1∼0.2 randomly)
Case 8	Gaussian (zero-mean, and noise variance is different for different bands and is selected from 0.1∼0.2 randomly) +salt and pepper(selected from 0 to 20% randomly) in all bands
Case 9	Gaussian (zero-mean, and noise variance is different for different bands and is selected from 0.1∼0.2 randomly) +salt and pepper(selected from 0 to 20% randomly) in all bands + dead-lines in 61∼70 bands (with the number of stripes randomly selected from 1 to 10 and the width of the dead-lines randomly generated from 1 to 3)

Extracting all items containing $\mathcal{V}$ from (37), similar to the process of solving $\mathcal{V}$ in the HSI denoising model, the detail of the l_0-l_1HTV method for HSI CS is summarized in Algorithm 2.

VI. EXPERIMENTAL RESULTS AND DISCUSSION

In this section, extensive experiments on HSI denoising and HSI compressed sensing are conducted to evaluate the performances of the proposed method. In the denoising experiments, we select four types of methods: TV based method, i.e., 2DTV, ASSTV [26], SSTV [27], HSSTV [28], E-3DTV [39], and LR-based method, i.e., LRMR [29]. In the CS experiments, l_0-l_1HTV is compared with including 2DTV, ASSTV [26], SSTV [27], and LR-based method, i.e., LR and joint-sparse recovery (SparCS) [55]. Three TV norms are replaced in (35) and also solved by ADMM.

In simulated experiments, we employ the mean peak signal-to-noise ratio (MPSNR), the mean structural similarity index (MSSIM), the Erreur Relative Globale Adimensionnelle de Synthese (ERGAS), and the mean spectral angle distance (MSAD) to evaluate HSI restoration qualities as follows:

$$\text{MPSNR} = \frac{1}{z}\sum_{i=1}^{z}\text{PSNR}(\mathcal{X}(:,:,i), \overline{\mathcal{X}}(:,:,i)) \tag{46}$$

$$\text{MSSIM} = \frac{1}{z}\sum_{i=1}^{z}\text{SSIM}(\mathcal{X}(:,:,i), \overline{\mathcal{X}}(:,:,i)) \tag{47}$$

$$\text{ERGAS} = \sqrt{\frac{1}{z}\sum_{i=1}^{z}\frac{\text{mse}\big(\mathcal{X}(:,:,i), \overline{\mathcal{X}}(:,:,i)\big)}{\text{Mean}_2(\mathcal{X}(:,:,i))}} \tag{48}$$

$$\text{MSAD} = \frac{1}{hv}\sum_{i=1}^{hv}\frac{180}{\pi}\times\arccos\frac{\big(\mathbf{X}_{(3)}(i,:)\big)^T\cdot\big(\overline{\mathbf{X}}_{(3)}(i,:)\big)}{\big\|\mathbf{X}_{(3)}(i,:)\big\|\cdot\big\|\overline{\mathbf{X}}_{(3)}(i,:)\big\|} \tag{49}$$

where $\mathcal{X}(:,:,i)$ and $\overline{\mathcal{X}}(:,:,i)$ denote the ith band of the reference and recovered images, and $\mathbf{X}_{(3)}(i,:)$ and $\overline{\mathbf{X}}_{(3)}(i,:)$ are the ith spectral signatures of two images.

A. HSI Mixed Noise Removal Experiments

Two HSI data sets are adopted including the Washington dc Mall (WDC) data set and the Indian synthetic (Indian-s) data. The WDC data set was captured by the HS digital imagery collection experiment. A subregion image in the size of $200 \times 200 \times 160$ holds complex structure and texture information. The Indian synthetic (Indian-s) data has been considered in [29], [33], [38], which is generated by USGS splib06 using the ground truth of the Indian Pines data set. The Indian-s data contain 145×145 pixels and 175 bands, with evident local smoothness information.

The noise of HSI data generally manifests as a mixture of Gaussian noise, salt and pepper noise, deadline noise, and strips. The details of noise cases are given in Table I. Three different kinds of the Gaussian noise are taken into account. In cases 1∼3, the Gaussian noise is randomly added to all bands of HSIs. The noisy intensity of different bands is different, and the SNR values of the Gaussian noise vary from 10∼20 dB randomly. For cases 4∼6, the variance σ_z^2 of the Gaussian noise is defined as follows [46]:

$$\sigma_z^2 = \delta^2\frac{\exp\{-(z-Z/2)^2/(2\eta^2)\}}{\sum_{z=1}^{Z}\exp\{-(z-Z/2)^2/(2\eta^2)\}} \tag{50}$$

where δ and η are two adjustable parameters, and the mean noise SNR is set as 12.12 dB.

1) WDC Data Set: In terms of visual quality, Fig. 2 shows band 70 of WDC data denoised by eight different methods under Case 5. ASSTV fails to remove all noise completely. Although noise is largely suppressed by 2DTV, its result yields some blurry areas and loses much detailed information. HSSTV and E-3DTV perform comparatively better than SSTV and LRMR in mixed noise removal, but their results exist more or less undesired artifacts in the green rectangle area. As observed, the proposed l_0-l_1HTV produces superior denoised results without any oversmoothing or artifacts and, meanwhile, preserves clearer structure and texture information, such as four vertical boundaries in the middle of the recovered image.

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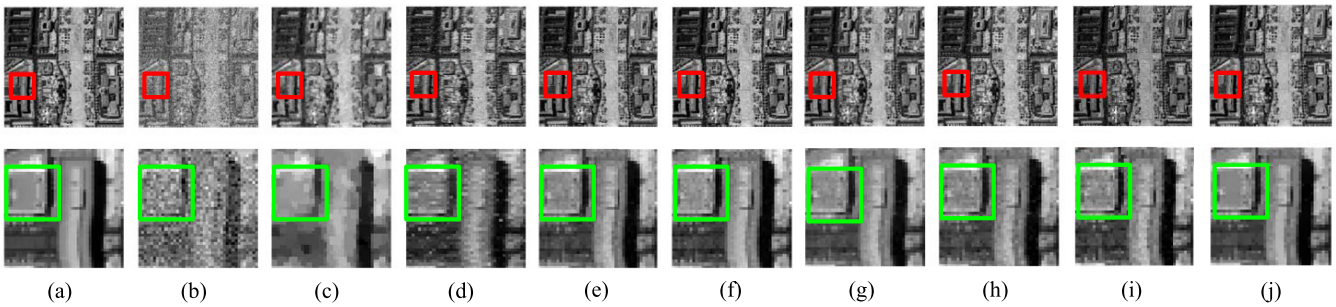


Fig. 2. Denoised results by all the compared methods for WDC data set. (a) Original band 70. (b) Simulated noise band of case 5. (c) 2DTV. (d) ASSTV. (e) SSTV. (f) LRMR. (g) HSSTV1. (h) HSSTV2. (i) E-3DTV. (j) l_0-l_1HTV.

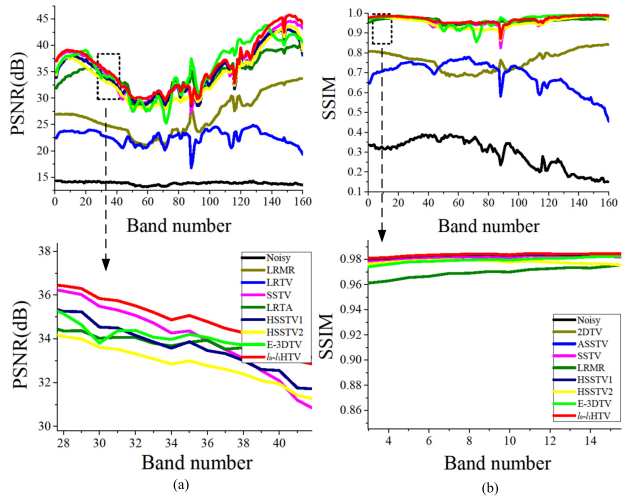


Fig. 3. (a) PSNR and (b) SSIM values of each band in Case 5 for the WDC data set.

To quantitatively compare the performances of these methods, we list the assessment results in Table II, with the best ones marked in bold and the second ones underlined. Consistent with visual comparison, the proposed method achieves the highest evaluation scores among seven denoised approaches. SSTV, HSSTV, and E-3DTV outperform LRMR under cases 1∼6 since these state-of-the-art TV-based methods ponder more local smoothness information. E-3DTV and LRMR have similar denoised results under cases 7∼9. Nevertheless, LRMR explores the LR property of HSI and enables to handle complex Gaussian noise, leading to higher MSSIM values than HSSTV and E-3DTV. As ASSTV utilizes local information along with three directions of HSI, ASSTV gains higher scores than 2DTV. Fig. 3 presents the PSNR and SSIM values of each band in Case 5 for the WDC data set. Due to the same curves of l_0-l_1HTV, LRMR, HSSTV, and E-3DTV, the curves in the black dashed frames are enlarged. It is obvious that the proposed method shown in red provides the best performances.

2) Indian-s Data Set: Fig. 4 shows the denoised performances of compared methods under Case 9. As shown in Fig. 4(a), Indian-s data has much sharper edges than the WDC data set. Similarly, ASSTV still remains the amount of noise, and 2DTV generates serious oversmoothing. ASSTV and SSTV fail to remove strips completely. Unlike SSTV, LRMR restores a better image without lots of artifacts since LRMR exploits LR prior information. HSSTV compensates for artifacts of a pure SSTV algorithm to a certain extent. However, HSSTV just exploits the local differences of HSI, which causes wrong boundary judgment. Although E-3DTV outperforms the abovementioned algorithms, there exist fewer artifacts in the smoothing areas since E-3DTV only utilizes the local spatial information and the sparsity insight of 3DTV. The white area on the left reveals that l_0-l_1HTV obtains the best smoothness and the sharpest edges. l_0-l_1HTV is superior to the other approaches.

TABLE II
QUANTITATIVE RESULTS OF ALL THE METHODS UNDER DIFFERENT MIXED NOISE CASES FOR WDC DATA SET

Noise Case	WDC Dataset									
	Index	Noisy	2DTV	ASSTV	SSTV	LRMR	HSSTV1	HSSTV2	E-3DTV	l_0-l_1HTV
Case 1	MPSNR	27.22	29.96	32.95	37.84	36.85	38.30	38.09	38.18	**38.89**
	MSSIM	0.7973	0.8779	0.9476	0.9776	0.9797	0.9809	0.9794	0.9819	**0.9834**
	EGRAS	11.9726	7.6773	5.5183	3.3347	3.2881	2.9490	3.0544	3.0633	**2.7484**
	MSAD	10.6152	7.5971	4.7666	3.1015	3.6719	2.8716	2.9465	3.0112	**2.6935**
Case 2	MPSNR	14.11	26.14	22.76	37.15	36.00	37.17	36.71	38.13	**38.27**
	MSSIM	0.3140	0.7781	0.7005	0.9767	0.9752	0.9776	0.9763	0.9804	**0.9813**
	EGRAS	46.9504	11.1352	17.8865	3.5510	3.6063	3.2904	3.4705	3.1541	**2.9667**
	MSAD	36.9857	11.6210	14.4817	3.3381	4.0624	3.2414	3.4029	3.0299	**2.9030**
Case 3	MPSNR	11.30	25.04	19.85	36.04	35.15	33.16	33.53	36.06	**36.82**
	MSSIM	0.1756	0.6995	0.5508	0.9689	0.9679	0.9575	0.9564	0.9702	**0.9750**
	EGRAS	46.9504	12.7272	25.3070	3.7629	4.1288	4.8508	4.7852	3.9624	**3.4041**
	MSAD	43.6584	13.3058	19.1071	3.7519	4.6473	4.7344	4.7852	3.8913	**3.4023**
Case 4	MPSNR	24.34	27.40	31.26	36.40	35.13	36.35	36.86	36.19	**37.00**
	MSSIM	0.7008	0.8029	0.9099	0.9674	0.9689	0.9713	0.9747	0.9698	**0.9734**
	EGRAS	15.2816	10.3531	6.9993	3.8969	3.6063	3.6474	3.6205	3.8483	**3.5670**
	MSAD	14.1859	10.0657	6.1004	3.6533	4.0624	3.5669	3.4223	3.7338	**3.3771**
Case 5	MPSNR	13.85	25.73	22.60	35.58	34.24	35.60	34.32	35.89	**36.25**
	MSSIM	0.2919	0.7277	0.6873	0.9622	0.9618	0.9632	0.9519	0.9655	**0.9670**
	EGRAS	47.7630	11.8439	18.0731	4.1833	4.4109	4.2951	4.7135	4.0372	**3.8344**
	MSAD	37.6946	12.1861	14.8037	3.9773	4.9412	4.1192	4.5168	3.8409	**3.6591**
Case 6	MPSNR	11.16	24.56	19.73	34.52	33.28	32.50	32.08	33.67	**35.03**
	MSSIM	0.1662	0.6796	0.5383	0.9541	0.9539	0.9428	0.9376	0.9481	**0.9605**
	EGRAS	65.9574	13.4884	25.4364	4.6480	4.9389	5.5669	5.8622	5.7319	**4.3581**
	MSAD	44.1772	14.0139	19.4128	4.4672	5.4491	5.3261	5.6505	5.2271	**4.1878**
Case 7	MPSNR	16.48	20.43	25.67	30.10	31.17	31.27	29.43	31.14	**31.38**
	MSSIM	0.3543	0.5043	0.8107	0.8853	0.9178	0.9170	0.8741	0.9049	**0.9198**
	EGRAS	34.9656	23.1351	12.3470	7.3824	6.2509	6.2464	7.7847	6.6029	**6.2090**
	MSAD	32.4990	22.9750	10.5510	7.9050	7.2501	6.6695	8.3656	6.9146	**6.6289**
Case 8	MPSNR	12.58	15.39	22.20	29.41	29.68	29.00	28.92	29.71	**30.07**
	MSSIM	0.2220	0.3104	0.6337	0.8708	**0.9054**	0.8825	0.8785	0.8726	0.8951
	EGRAS	58.4574	43.0059	18.9829	8.0110	7.4304	8.2410	8.3480	8.9605	**7.3081**
	MSAD	41.8212	34.0091	15.9943	8.4385	8.2500	8.2950	8.4206	8.3783	**7.5561**
Case 9	MPSNR	12.50	15.30	22.14	29.30	29.57	28.50	28.35	29.63	**30.00**
	MSSIM	0.2199	0.3763	0.6315	0.8160	**0.9033**	0.8600	0.8554	0.8632	0.8949
	EGRAS	58.3482	42.5358	18.8190	7.9522	7.5299	8.9066	9.0515	9.5305	**7.2896**
	MSAD	42.1183	33.6726	16.0386	8.4101	8.3486	8.8264	8.9889	8.6956	**7.5511**

Table II gives the quantitative results of eight denoised algorithms and further proves the best performances of our

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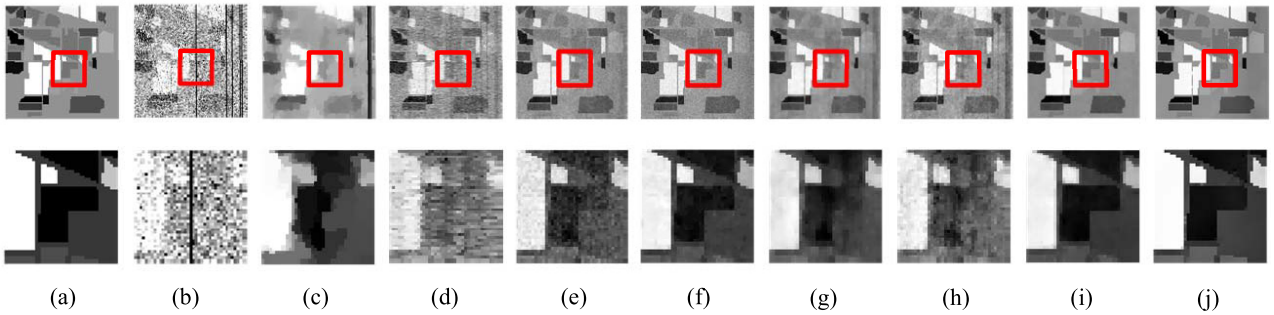


Fig. 4. Denoised results by all the compared methods for Indian-s data set. (a) Original band 65. (b) Simulated noise band of case 9. (c) 2DTV. (d) ASSTV. (e) SSTV. (f) LRMR. (g) HSSTV1. (h) HSSTV2. (i) E-3DTV. (j) l_0-l_1HTV.

TABLE III
QUANTITATIVE RESULTS OF ALL THE METHODS UNDER DIFFERENT MIXED NOISE CASES FOR INDIAN-S DATA SET

Noise Case	Indian-s Dataset									
	Index	Noisy	2DTV	ASSTV	SSTV	LRMR	HSSTV1	HSSTV2	E-3DTV	l_0-l_1HTV
Case 1	MPSNR	26.34	31.80	34.04	42.11	41.50	42.20	40.84	42.09	**48.36**
	MSSIM	0.5125	0.9092	0.7987	0.9706	0.9554	0.9853	0.9723	0.9864	**0.9976**
	ERGAS	11.4981	5.3534	5.1290	1.6352	1.7970	1.6658	1.9309	2.3059	**0.8380**
	MSAD	1.8373	5.6082	4.0230	1.7533	1.8373	1.7766	2.0184	1.6364	**0.8329**
Case 2	MPSNR	14.10	30.55	25.89	40.76	40.18	40.92	39.17	41.73	**46.86**
	MSSIM	0.1416	0.8869	0.6865	0.9457	0.9264	0.9775	0.9506	0.9882	**0.9967**
	ERGAS	45.2641	6.2247	12.7398	1.8873	2.0617	1.8495	2.3058	1.9456	**0.9965**
	MSAD	34.4532	6.5157	8.9093	2.0229	2.1259	1.8881	2.2816	1.7073	**0.9866**
Case 3	MPSNR	11.38	29.36	22.10	39.33	38.89	38.36	36.12	40.63	**45.50**
	MSSIM	0.0734	0.8613	0.5166	0.9457	0.9264	0.9676	0.9142	0.9843	**0.9958**
	ERGAS	63.1225	7.1611	20.3016	2.2232	2.4029	2.4883	3.2396	2.2755	**1.0183**
	MSAD	39.9088	7.4905	12.3116	2.3942	2.4565	2.4307	3.1494	1.9189	**1.0323**
Case 4	MPSNR	23.51	30.79	30.83	39.17	38.84	40.78	39.97	40.07	**45.89**
	MSSIM	0.4147	0.8796	0.6741	0.9682	0.9081	0.9726	0.9317	0.9742	**0.9974**
	ERGAS	16.0971	6.0026	8.1258	2.3333	2.5499	1.9414	2.5508	2.6449	**1.1403**
	MSAD	14.6163	6.1425	6.0690	2.3942	2.4827	1.9183	2.3623	2.0342	**1.0600**
Case 5	MPSNR	13.62	29.78	25.25	38.14	37.91	38.95	36.76	39.59	**44.69**
	MSSIM	0.1264	0.8606	0.6348	0.9075	0.8965	0.9632	0.8987	0.9739	**0.9940**
	ERGAS	16.0887	6.7552	13.1283	2.6097	2.8033	2.3932	3.2468	2.9820	**1.2872**
	MSAD	36.5602	6.9482	9.3958	2.6529	2.7355	2.3290	3.0067	2.1509	**1.2322**
Case 6	MPSNR	11.13	28.40	21.67	36.81	36.56	36.54	34.09	37.93	**42.85**
	MSSIM	0.0681	0.8275	0.4806	0.8953	0.8715	0.9458	0.8492	0.9695	**0.9902**
	ERGAS	46.3208	7.9260	20.6494	3.0334	3.4358	3.1618	4.3455	3.0903	**1.6262**
	MSAD	35.5602	8.1290	12.7695	3.0798	3.1478	2.9339	3.9652	2.5740	**1.5174**
Case 7	MPSNR	16.07	28.40	27.64	32.04	34.66	36.09	32.53	39.64	**40.25**
	MSSIM	0.1499	0.7921	0.6297	0.7656	0.8819	0.9468	0.8385	0.9804	**0.9880**
	ERGAS	36.0327	8.3164	9.9599	5.9355	3.7659	3.4308	5.1724	2.5135	**2.1761**
	MSAD	32.9956	9.2680	9.2830	6.6732	4.2082	3.8704	5.8814	**2.3930**	2.4115
Case 8	MPSNR	12.93	27.46	23.17	31.32	33.86	33.19	30.08	36.92	**39.37**
	MSSIM	0.0956	0.7982	0.5024	0.7403	0.8631	0.9231	0.7656	0.9724	**0.9838**
	ERGAS	54.4352	9.0489	18.7659	6.3556	4.1185	5.3218	7.2719	3.2275	**2.4493**
	MSAD	36.5381	10.0114	12.2540	7.0768	4.5191	5.1050	7.2911	3.1632	**2.6785**
Case 9	MPSNR	12.87	27.19	23.10	31.22	33.78	32.96	29.87	36.80	**39.28**
	MSSIM	0.0951	0.7942	0.5014	0.7403	0.8615	0.9208	0.7603	0.9720	**0.9852**
	ERGAS	54.5384	9.2997	18.8065	6.3824	4.1608	5.3789	7.3462	3.2776	**2.4148**
	MSAD	36.5962	10.1866	12.2709	7.1077	4.5275	5.1436	7.3547	3.1931	**2.6458**

proposed l_0-l_1HTV. For this kind of data set with sharper edges, E-3DTV and HSSTV bring better results than 2DTV, ASSTV, and LRMR in most of the cases. E-3DTV obtains the second-highest metric values among seven comparing methods. HSSTV deals with complex Gaussian noise cases, such as cases 8 and 9, a little worse than LRMR. Due to the good use of local HSI information, HSSTV obtains better MSSIM scores than LRMR. Meanwhile, as we can see in Fig. 5, the PSNR and SSIM values of each band obtained by the proposed l_0-l_1HTV are much higher than the other state-of-the-art algorithms.

3) Real-World Data Set: The Indian Pines data set was acquired by the NASA AVIRIS instrument in Northwestern Indiana, containing 145 × 145 and 224 spectral reflectance bands. Fig. 6 shows that the original band 220 of this scene is corrupted by mixed noise, and target objects can barely be recognized, which directly hinders further HSI analysis.

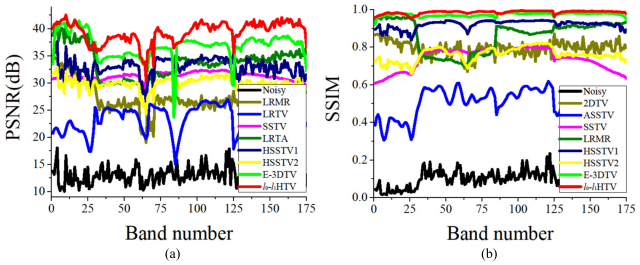


Fig. 5. (a) PSNR and (b) SSIM values of each band in case 9 for Indian-s data set.

The above eight methods are implemented for HSI denoising. 2DTV and ASSTV lose their utility when they encounter heavy noise cases. SSTV and LRMR remove lots of noise, but they produce artifacts. The result denoised by E-3DTV appears blurred image edges and fewer artifacts. In contrast, HSSTV and l_0-l_1HTV restore the clearest image without artifacts.

In real data experiments, the above image assessments are useless since the reference images are unknown. Meanwhile, preserving sharp image edges benefits further HSI tasks, such as classification. Therefore, the supervised classification experiment is designed, and the well-known support vector machines (SVM) [56]–[58] is applied to quantitatively compare the performances of different methods. The first and second columns of Table IV show 16 different classes and the number of training (random 10%) and test samples. The accuracy for each class, kappa coefficient, overall accuracy (OA), and average accuracy (AA) are depicted in Table IV, and eight denoising techniques can be quantitatively evaluated. Fig. 7 presents the visual comparison of classification results before and after denoising. It is conspicuously seen that classification accuracies are significantly fortified after the denoising process. Due to the oversmoothing phenomenon, the results by 2DTV and ASSTV appear large misclassified areas. Therefore, their kappa, OA, and AA scores have risen less. E-3DTV presents higher classification accuracies than LRMR since E-3DTV explores the sparsity of subspace on the gradient maps and is one of the advanced 3DTV variant algorithms. SSTV provides higher metric values, but artifacts cause incorrect classification results for each class. As kappa, OA, and AA show, two HSSTV methods have similar classification results, which are a little lower than the proposed method. l_0-l_1HTV produces the highest evaluation scores, with kappa, OA, and AA of 0.9594, 0.9644, and 0.9666, respectively.

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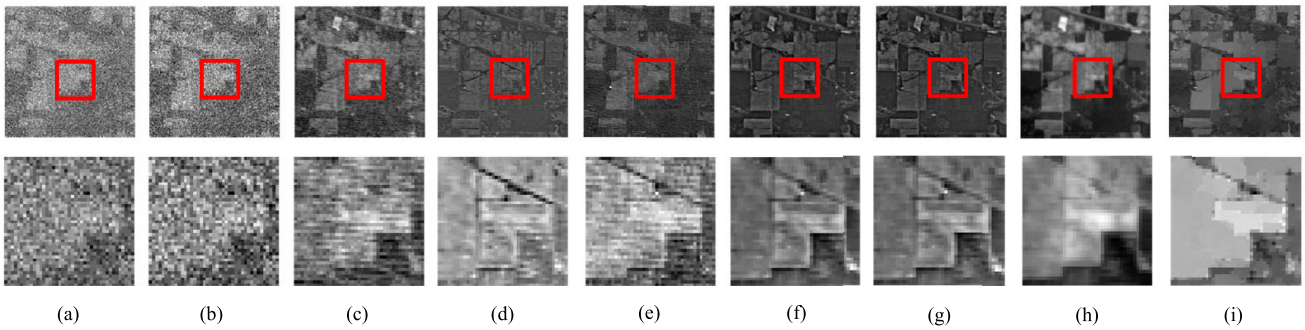


Fig. 6. Denoised results by all the compared methods on the Indian Pine data. (a) Original band 220. (b) 2DTV. (c) ASSTV. (d) SSTV. (e) LRMR. (f) HSSTV1. (g) HSSTV2. (h) E-3DTV. (i) l_0-l_1HTV.

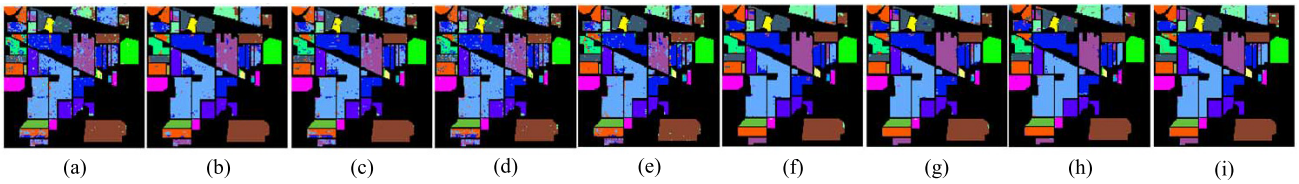


Fig. 7. Classification results by all the compared methods in the Indian Pine data. (a) Original. (b) 2DTV. (c) ASSTV. (d) SSTV. (e) LRMR. (f) HSSTV1. (g) HSSTV2. (h) E-3DTV. (i) l_0-l_1HTV.

TABLE IV
CLASSIFICATION RESULTS ON INDIAN PINES DATA BY DIFFERENT METHODS

Class	Train/Test	Noisy	2DTV	ASSTV	SSTV	LRMR	HSSTV1	HSSTV2	E-3DTV	$l_0 - l_1$HTV
Alfalfa	5/41	0.6341	0.6585	0.7073	0.7804	0.7317	0.8780	0.8537	0.6585	**0.9512**
Corn-notill	143/1285	0.8202	0.9253	0.8677	0.9167	0.9105	0.9183	0.9198	0.8996	**0.9447**
Corn-mintill	83/747	0.7161	0.9076	0.7778	0.9518	0.8955	0.9572	**0.9625**	0.8983	0.9394
Corn	24/213	0.8028	**0.9848**	0.8967	0.9154	0.9248	0.9765	0.9390	0.9577	0.9437
Grass-pasture	49/434	0.9377	**0.9931**	0.9470	0.9723	0.9354	0.9700	0.9677	0.9493	0.9816
Grass-trees	73/657	0.9497	0.9833	0.9696	0.9863	0.9619	0.9897	0.9848	0.9833	**0.9985**
Grass-pasture-mowed	3/25	0.6400	0.7200	0.7600	0.8400	0.6400	0.8800	0.800	0.5600	**0.9600**
Hay-windrowed	48/430	0.9906	0.9930	**1.0000**	0.9953	0.9976	0.9977	0.9977	0.9977	**1.0000**
Oats	2/18	0.7222	0.9444	**1.0000**	0.9444	0.8333	0.9359	0.9444	**1.0000**	**1.0000**
Soybean-notill	98/874	0.7059	0.8272	0.7780	**0.9405**	0.8821	0.9359	0.9176	0.8833	0.9245
Soybean-mintill	246/2209	0.8415	0.9140	0.8873	0.9601	0.9388	0.9520	0.9579	0.9511	**0.9765**
Soybean-clean	60/533	0.8480	0.9587	0.9043	0.9699	0.9324	**0.9737**	0.9625	0.9043	0.9437
Wheat	21/184	0.9945	**0.9946**	**0.9946**	0.9945	0.9945	0.9891	0.9891	**0.9946**	0.9783
Woods	127/1138	0.9516	0.9824	0.9754	0.9674	0.9455	0.9736	0.9824	0.9807	**0.9851**
Buildings-Grass-Trees-Drives	39/347	0.6311	0.8559	0.6513	0.8587	0.8040	**0.9510**	0.9424	0.9049	0.9308
Stone-Steel-Towers	10/83	0.9518	**1.0000**	9518	0.9638	0.9759	**1.0000**	**1.0000**	**1.0000**	**1.0000**
kappa	-	0.8197	0.9197	0.8677	0.9451	0.9131	0.9510	0.9496	0.9271	**0.9594**
OA	-	0.8423	0.9293	0.8842	0.9518	0.9237	0.9569	0.9557	0.9360	**0.9644**
AA	-	0.8212	0.9192	0.8793	0.9349	0.8911	0.9588	0.9451	0.9077	**0.9666**

B. HSI Compressed Sensing Experiments

Three HSI data experiments are employed to validate the effectiveness of l_0-l_1HTV on HSI CS with seven different sample ratio, i.e., 0.05, 0.1, 0.2, 0.4, and 0.6. Three data sets include Pavia University, CAVE Toy, and Reno, and their cubes used for HSI CS experiments are of sizes $250 \times 121 \times 130$, $300 \times 300 \times 31$, and $150 \times 150 \times 100$.

1) Pavia University: In terms of visual quality, the representative bands of eight HSI CS results in the sampling ratio $r = 0.4$ are presented in Fig. 8. As shown in the green square of the enlarged areas, reconstructed images by 2DTV, ASSTV, and HSSTV are smoothed excessively, which causes the loss of street detailed information. As observed from the white areas, SSTV and SparCS lead to more or fewer artifacts. Meanwhile, HSSTV overcomes the artifacts of SSTV, and E-3DTV compensates for the defect of oversmoothing. l_0-l_1HTV provides the best result without any artifact, and the comparison of PSNR and SSIM values of all the bands in Fig. 9 also illustrates the superiority of our l_0-l_1HTV method.

Table V presents the quantitative indices of eight different sampling ratios on the Pavia university data set. SSTV and SparCS can reconstruct great images in high sampling ratios, but poor results are obtained in low sampling ratios. Nevertheless, the results of 2DTV are just the opposite. As two modified TV-based methods, HSSTV and E-3DTV always obtain better results than SSTV, but they still have quality decays with sample ratios decreasing. These indicate the instabilities of the above three methods. l_0-l_1HTV provides much

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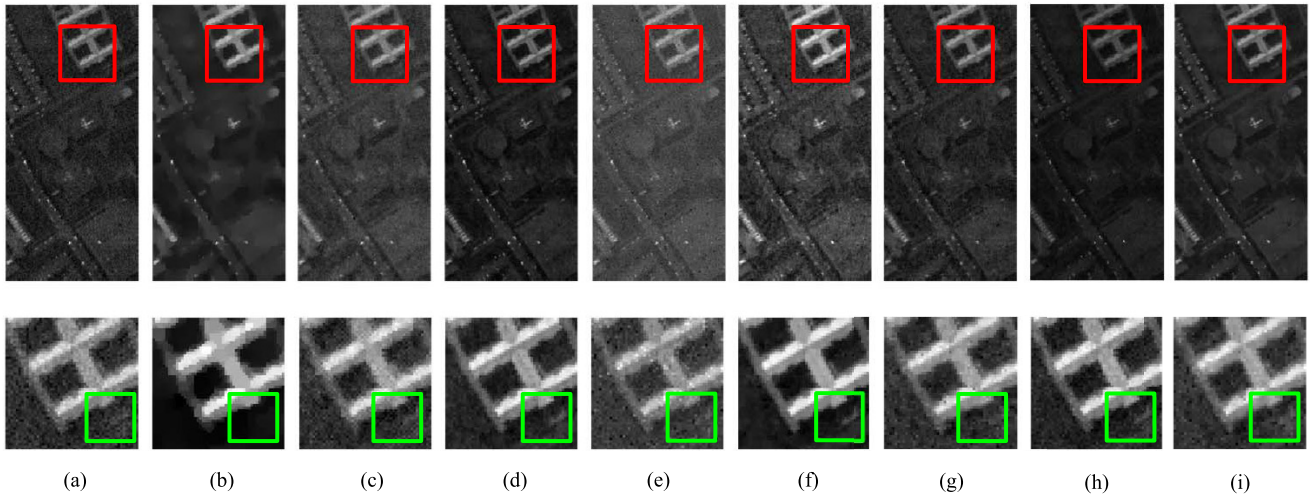


Fig. 8. HSI CS results on the Pavia University data set, and the sample ratio is 0.4. (a) Original band 1. (b) 2DTV. (c) ASSTV. (d) SSTV. (e) SparCS. (f) HSSTV1. (g) HSSTV2. (h) E-3DTV. (i) l_0-l_1HTV.

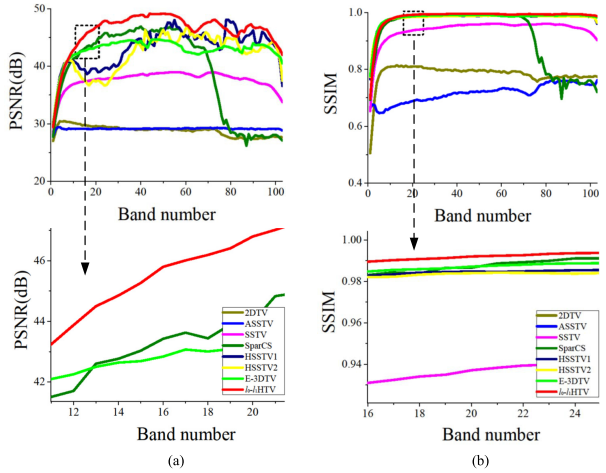


Fig. 9. (a) PSNR and (b) SSIM values of each band for Pavia University under $r = 0.4$.

TABLE V

QUANTITATIVE RESULTS OF ALL THE METHODS UNDER DIFFERENT CS CASES FOR THE PAVIA UNIVERSITY DATA SET

Ratio	Index	2DTV	ASSTV	SSTV	SparCS	HSSTV1	HSSTV2	E-3DTV	l_0-l_1HTV
		Pavia university Dataset							
0.05	MPSNR	21.48	20.37	21.17	19.71	25.15	22.20	26.89	**29.63**
	MSSIM	0.4587	0.3174	0.4446	0.3002	0.7009	0.6681	0.7508	**0.8492**
	ERGAS	20.0014	22.8979	20.9138	25.0856	13.3918	18.5314	10.8490	**7.8779**
	MSAD	23.8498	27.7240	25.0375	29.2799	14.6353	19.8596	11.7016	**8.8695**
0.1	MPSNR	23.43	22.04	23.01	20.83	28.57	27.93	30.39	**36.33**
	MSSIM	0.5374	0.4216	0.5270	0.3597	0.9036	0.8964	0.8613	**0.9411**
	ERGAS	15.8906	18.8693	16.9183	23.2884	9.2553	9.8463	7.2841	**3.7945**
	MSAD	18.7013	22.4843	19.9202	20.8280	10.2367	10.9056	7.6556	**4.2857**
0.2	MPSNR	25.69	25.57	27.79	26.19	36.49	32.09	34.24	**41.11**
	MSSIM	0.6531	0.5768	0.7175	0.6363	0.9666	0.9560	0.9390	**0.9717**
	ERGAS	12.2199	12.5174	9.7927	12.5922	4.0403	6.4183	4.7032	**2.4319**
	MSAD	14.2670	14.6661	11.3162	13.7806	4.3824	7.0270	5.3836	**2.5706**
0.4	MPSNR	28.80	29.13	37.60	39.23	43.11	42.14	42.82	**46.15**
	MSSIM	0.7841	0.7160	0.9389	0.9173	0.9803	0.9801	0.9826	**0.9844**
	ERGAS	8.5431	8.3255	3.3359	4.5526	2.1228	2.3508	1.9911	**1.6831**
	MSAD	9.8675	9.7278	3.7420	3.9402	2.1740	2.4210	2.1164	**1.5689**
0.6	MPSNR	31.15	34.69	45.58	47.04	47.25	46.32	46.81	**47.92**
	MSSIM	0.8533	0.8875	0.9836	0.9875	0.9858	0.9852	0.9909	**0.9881**
	ERGAS	6.5299	4.4228	1.7491	1.6358	**1.4127**	1.5741	1.3848	1.4497
	MSAD	7.5053	5.1355	1.6679	1.4617	1.3862	1.5693	1.3894	**1.3116**

higher metrics than the other approaches, with lower sampling ratios.

2) Toy: Compared with the above Toy HSIs, more regular clothes texture information exists in the magnified areas of Fig. 10(a). Unfortunately, ASSTV and SSTV fail to recover the final HSI under a low sampling ratio $r = 0.2$. Although a few texture details can be found in the results by 2DTV and SparCS, it is hard to distinguish the colors of texture information. As the zoomed areas of the yellow frame show, much white background information is lost by two HSSTV algorithms. E-3DTV produces a better background result, but there still exist few artifacts. The l_0-l_1HTV method provides the best result with reconstructing most information of the original image. The PSNR and SSIM values of each band in Case 9 for the Indian-s data set are shown in Fig. 9. We can observe that, in almost all bands, the metric values of our proposed method (the red curves) are higher than those of the other approaches.

Table VI lists the four metric values of different methods. Under almost all cases, the metric scores obtained by using l_0-l_1HTV are the highest among all the techniques. 2DTV, ASSTV, and SSTV outperform SparCS under most sampling ratios, which demonstrates that TV methods are more suitable for recovering HSIs with more texture information than pure LR methods. The MSSIM, ERGAS, and MSAD values of HSSTV are the highest when high sample ratio $r = 0.6$, but the metrics of HSSTV decrease more than E-3DTV and l_0-l_1HTV with decreasing sample ratios.

3) Reno Data Set: Fig. 12 depicts the reconstructed results of band 2 of the Reno data set by different methods, with the random sample ratio $r = 0.1$. 2DTV causes spatial oversmoothing. SparCS exploits the LR property of HSI, but SparCS ignores spatial smoothness, which leads to the lack of local structure information. Due to the existence of artifacts, blurry image edges can hardly be distinguished in the result recovered by SSTV. Based on local spatial and spatiospectral information, HSSTV recovers some clear image edges but loses lots of pixels in the white areas. It is evident that E-3DTV and l_0-l_1HTV get the best reconstruction, but l_0-l_1HTV recovers the complete white areas and the sharpest edges. The PSNR and SSIM of each band restored by different approaches are shown in Fig. 13, and the results demonstrate that l_0-l_1HTV marked in red achieved the best results in all the bands.

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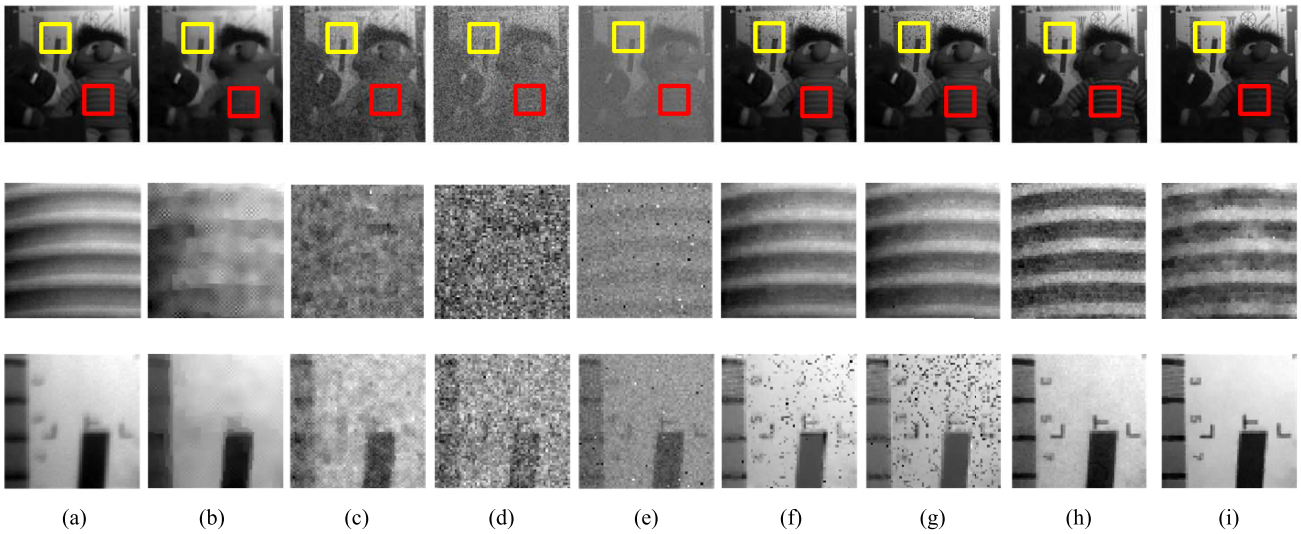


Fig. 10. HSI CS results on the Toy data set and sample ratio is 0.2. (a) Original band 2. (b) 2DTV. (c) ASSTV. (d) SSTV. (e) SparCS. (f) HSSTV1. (g) HSSTV2. (h)E-3DTV. (i) l_0-l_1HTV.

TABLE VI
QUANTITATIVE RESULTS OF ALL THE METHODS UNDER DIFFERENT CS CASES FOR THE TOY DATA SET

Ratio	Toy Dataset								
	Index	2DTV	ASSTV	SSTV	SparCS	HSSTV1	HSSTV2	E-3DTV	l_0-l_1HTV
0.05	MPSNR	13.18	12.39	12.91	10.89	14.62	13.36	20.42	**23.74**
	MSSIM	0.2033	0.1415	0.1299	0.0757	0.3027	0.6318	0.3896	**0.6234**
	ERGAS	47.0874	51.5439	48.4195	61.4419	39.7529	45.9325	20.5011	**13.8817**
	MSAD	39.2882	44.2176	40.7690	53.3116	27.8701	33.6779	10.7912	**16.6667**
0.1	MPSNR	15.83	13.78	14.51	12.82	18.55	16.10	27.66	**32.23**
	MSSIM	0.2746	0.1716	0.1572	0.1091	0.8110	0.7610	0.7519	**0.8800**
	ERGAS	34.8259	44.0011	40.2711	49.3815	25.2887	33.5346	8.8715	**5.2783**
	MSAD	27.0173	36.0451	32.6395	41.5952	17.8904	24.3612	6.47281	**4.0486**
0.2	MPSNR	26.30	19.65	18.62	19.11	29.60	25.42	33.85	**36.55**
	MSSIM	0.7171	0.3640	0.2635	0.2877	0.8326	0.7993	0.8957	**0.9462**
	ERGAS	10.6142	22.4768	25.2059	24.5570	7.0920	11.4444	4.4208	**3.3489**
	MSAD	7.5504	16.1117	19.5294	41.5952	5.4126	8.6127	3.2960	**2.5473**
0.4	MPSNR	37.23	31.63	31.15	30.6924	39.84	38.26	40.52	**41.23**
	MSSIM	0.9480	0.7557	0.7128	0.7133	0.9842	0.9803	0.9673	**0.9811**
	ERGAS	3.0369	5.6096	6.0629	89464	2.2174	2.6428	2.1044	**1.6689**
	MSAD	2.2502	4.2858	4.6485	5.4921	1.7069	2.0394	1.5899	**1.2115**
0.6	MPSNR	43.85	33.97	44.60	35.90	45.27	45.01	44.18	**47.92**
	MSSIM	0.9884	0.8182	0.9816	0.8457	**0.9894**	0.9891	0.9822	0.9881
	ERGAS	1.3753	4.2852	1.3168	5.0260	**1.1959**	1.1866	1.4131	1.4497
	MSAD	1.0637	3.3165	1.0021	3.1134	0.9190	**0.9015**	1.0589	1.3116

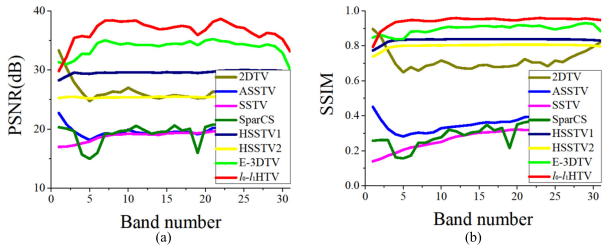


Fig. 11. (a) PSNR and (b) SSIM values of each band for Toy under $r = 0.2$.

Table VII provides the quantitative results of eight different sampling ratios on the Reno data set in terms of four evaluation indices. It can be easily seen that the proposed l_0-l_1HTV gives a significantly fortified performance under five sampling ratios compared with other competing methods. Since important structure details of the final HSIs are barely reconstructed, 2DTV and ASSTV get similar assessment scores. By using more local differences, E-3DTV and HSSTV recover the second-best and third-best results, respectively. Especially, E-3DTV is more beneficial for the CS recovery with lower sampling ratios. However, l_0-l_1HTV achieves the best reconstructions under all sampling ratios.

In summary, the structure information of the three data sets is different, while our proposed method always obtains the best-recovered images. l_0-l_1HTV can be a useful regularizer to ameliorate the performances for HSI denoising and CS tasks.

C. Discussion

1) Convergence of the Proposed Methods: The convergence is verified on HSI denoising and CS experiment data sets. Fig. 14(a) displays the RalCha of two denoised results for WDC and Indian-s data sets under noisy cases 5 and 9, respectively. RalCha is the relative change of two restored results, i.e., $||\mathcal{X}(i+1) - \mathcal{X}(i)||_F^2 / ||\mathcal{X}(i)||_F^2$, where $\mathcal{X}(i)$ is the ith recovered HSI image. Fig. 14(b) shows the RalCha of two recovered results for PaviaUA and Toy data sets under sampling ratio $r = 0.1$ and $r = 0.2$, respectively. It can be spotted that, as the iterations are conducted, the RalCha values of l_0-l_1HTV for HSI denoising and CS reconstruction monotonically decrease, finally approaching zero.

2) Parameter Analysis: In the l_0-l_1HTV model, four parameters should be carefully identified for the HSI denoising and CS. We analyze the impact of each parameter on the restoration results and discuss how to choose such parameters in our experiments. The denoised results are based on the simulated experiments in WDC and Indian-s data sets. The effect of each parameter on the CS is discussed under sampling ratio $r = 0.1$.

Parameter γ plays an important role to control the number of l_0 gradients. Due to the relationship with an HSI size, γ is replaced by the parameter γ' that is set to various percentages of total HSI size in Figs. 15(a) and 16(a). γ' is chosen from the set [0.1, 1.0] with 0.01 as step size. The Indian-s and Toy data sets have simpler texture information than the WDC and

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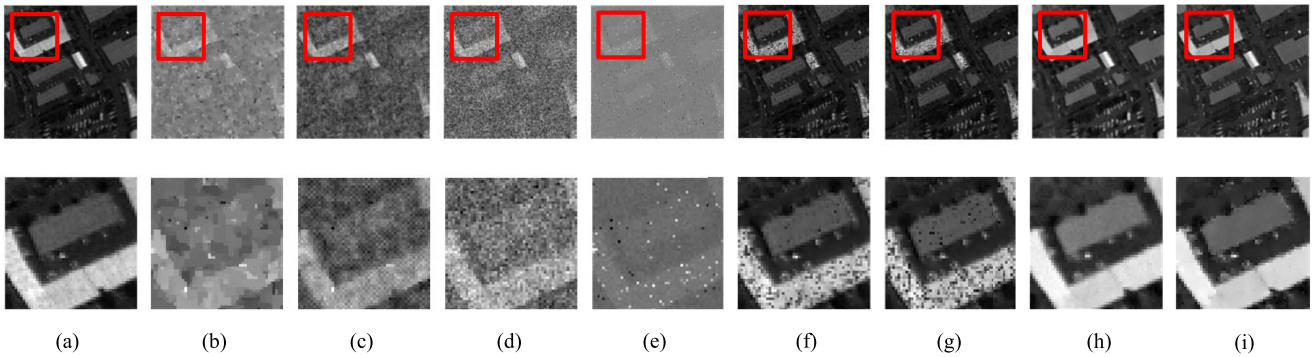


Fig. 12. HSI CS results on the Reno data set, and the sample ratio is 0.1. (a) Original band 2. (b) 2DTV. (c) ASSTV. (d) SSTV. (e) SparCS. (f) HSSTV1. (g) HSSTV2. (h) E-3DTV. (i) l_0-l_1HTV.

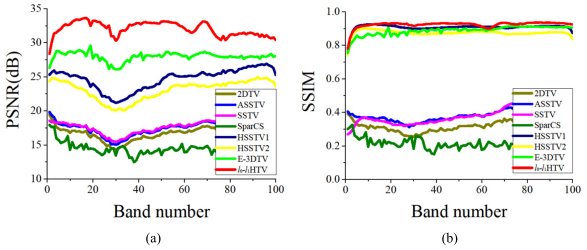


Fig. 13. (a) PSNR and (b) SSIM values of each band for the Reno data set under $r = 0.1$.

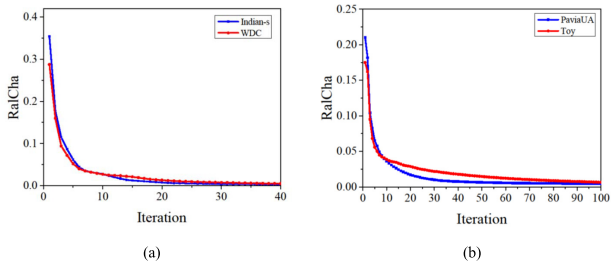


Fig. 14. RelCha values with respect to the number of iterations. (a) HSI denoising. (b) HSI CS.

TABLE VII

QUANTITATIVE RESULTS OF ALL THE METHODS UNDER DIFFERENT CS CASES FOR THE RENO DATA SET

Ratio	Reno Dataset								
	Index	2DTV	ASSTV	SSTV	SparCS	HSSTV1	HSSTV2	E-3DTV	l_0-l_1HTV
0.05	MPSNR	15.76	15.96	16.21	13.56	18.55	17.04	22.80	**25.11**
	MSSIM	0.2574	0.2743	0.2891	0.1488	0.7336	0.6472	0.6073	**0.7724**
	ERGAS	33.2416	32.5009	31.5179	45.2039	24.0110	28.4047	14.5842	**11.1754**
	MSAD	32.1112	31.2797	30.2264	41.6306	20.7043	24.7868	13.3769	**9.8166**
0.1	MPSNR	16.87	17.62	17.77	14.48	24.67	23.08	27.98	**32.05**
	MSSIM	0.3240	0.3867	0.3866	0.2189	0.9070	0.8743	0.8876	**0.9230**
	ERGAS	29.2555	26.8520	26.3792	38.1143	12.1361	14.4872	8.0589	**5.0768**
	MSAD	27.7898	25.1334	24.8107	14.4849	10.5626	12.6363	6.5857	**4.6492**
0.2	MPSNR	20.10	22.98	21.84	22.18	35.54	32.76	35.72	**38.94**
	MSSIM	0.5054	0.6348	0.5921	0.6234	0.9782	0.9744	0.9733	**0.9792**
	ERGAS	20.1403	14.3473	16.5678	16.5344	3.5288	5.0153	3.3380	**2.3849**
	MSAD	18.4749	12.8383	15.1398	14.8583	3.1360	4.3568	2.9699	**2.1355**
0.4	MPSNR	29.54	30.77	36.19	36.84	38.11	36.60	42.82	**45.81**
	MSSIM	0.8679	0.8669	0.9537	0.9306	0.9887	0.9858	0.9914	**0.9961**
	ERGAS	6.7873	5.8024	3.2823	5.1925	2.8389	3.3429	1.5316	**1.1066**
	MSAD	6.1679	5.3277	2.9512	3.6990	2.4193	2.8556	1.3672	**0.9465**
0.6	MPSNR	33.83	33.69	47.64	47.92	48.34	47.41	47.78	**48.81**
	MSSIM	0.9413	0.9179	0.9957	0.9890	0.9974	0.9950	0.9964	**0.9969**
	ERGAS	4.1420	4.1573	0.9421	1.7419	0.9277	1.0385	0.9366	**0.8553**
	MSAD	3.7852	3.8306	0.8052	0.9994	0.7756	0.8616	0.8017	**0.7165**

Pavia data sets; thus, γ' values for the Indian-s and Toy data sets are lower than the others.

Parameter λ is related to the strength of sparse noise norm. l_0-l_1HTV is tested using different values of λ varied among [0.01,0.1]. It is obvious that l_0-l_1HTV produces an appealing result when λ lies in the range of [0.05, 0.06].

Parameter λ_{tv} restricts the influence of the SSTV regularization on denoising or CS effectiveness. Figs. 15(c) and 16(b) present the robustness of the propose method to the changes of λ_{tv}. When λ_{tv} is given within [0.07, 0.08] for denoising and [8, 10] for CS, respectively, PSNR stabilizes at a high value.

Parameter β is used for the ADMM multipliers. First, β is set as 0.1 for denoising and 0.01 for CS, respectively, and then tuned gradually. β is fixed as 0.3 and 0.05 for denoising and CS, respectively, which achieves the best performances.

3) Computational Complexity: The computational complexities of the proposed algorithm for two HSI restorations are discussed in this section, and the first three parts for denoising and CS processes are universal as follows.

1) Solving the subproblem (22), (23), or (39), (40) includes two thresholds and two SSTV operators, which takes about $O(2hvz + 2hvz)$ iterations.
2) For (24) or (41), the sorting of the L2-norms of hv subvectors spends the most time and requires $O(hv(\log hv))$ time.
3) Most of the computation time for updating the multipliers lies on the SSTV and l_0HTV operators, which requires about $O(3hvz)$ iterations.

For HSI denoising, the threshold for solving $\mathcal{S}$-related problem requires $O(hvz)$, and the computation of LSQR algorithm used to update $\mathbf{x}$ costs $O(4hv + 5z)$.

For HSI CS, the LSQR algorithm is also utilized to update $\mathbf{x}$, and its computation costs $O(4hv + hvz)$.

Therefore, the computational complexities of the HSI denoising process and CS the process are in total $O(8hvz + 5z + hv(\log hv))$ and $O(8hvz + 4hv + hv(\log hv))$, respectively.

The running times of different methods on HSI denoising and CS reconstruction experiments are reported in Table VIII. All the experiments are conducted in MATLAB 2018b on the PC with an Intel i7 CPU at 2.60 GHz and 32 GB of memory. For HSI denoising, ASSTV is the fastest one among all the comparing methods, but most of the mixed

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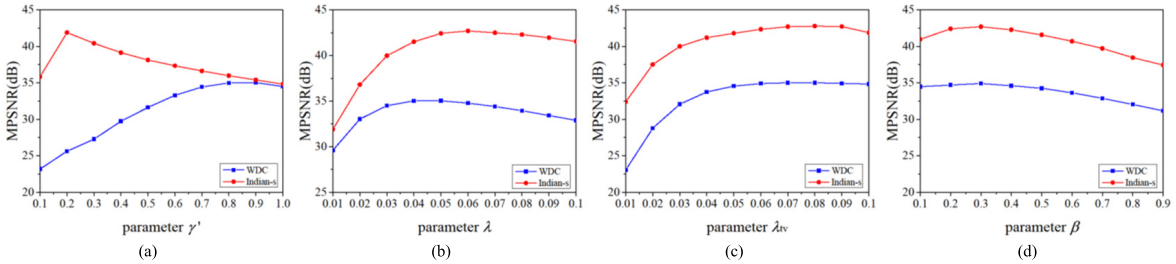


Fig. 15. MPSNR as a function of the parameters (i.e., (a) γ', (b) λ, (c) λ_{tv}, and (d) β) in the proposed method for HSI denoising on the WDC and Indian-s data sets.

TABLE VIII
RUNNING TIME COMPARISON OF DIFFERENT METHODS

HSI data sets	size	2DTV	ASSTV	SSTV	LRMR	HSSTV1	HSSTV2	E-3DTV	l_0-l_1HTV
WDC	$200 \times 200 \times 160$	314.5s	109.9s	564.3s	131.6s	1710.7s	1805.6s	390.5s	571.9s
Indian-s	$145 \times 145 \times 175$	189.9s	57.4s	432.3s	99.2s	1054.2s	1061.8s	193.10s	349.6s
Indian Pines	$145 \times 145 \times 220$	235.5s	72.5s	466.5s	115.1s	1124.6s	1286.3s	211.8s	400.6s
HSI data sets	size	2DTV	ASSTV	SSTV	SparCS	HSSTV1	HSSTV2	E-3DTV	l_0-l_1HTV
Pavia University	$250 \times 121 \times 103$	832.1s	1247.7s	941.3s	1516.9s	1126.1s	1274.9s	288.1s	778.3s
Toy	$300 \times 300 \times 31$	1025.7s	1303.7s	1066.2s	1355.1s	1334.9s	1465.4s	307.4s	819.7s
Reno	$150 \times 150 \times 100$	708.9s	1012.1s	734.7s	1027.5s	1028.4s	1135.9s	267.7s	679.4s

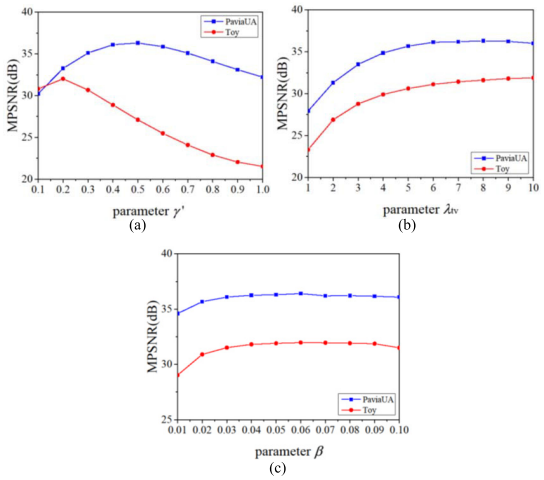


Fig. 16. MPSNR as a function of the parameters (i.e., (a) γ', (b) λ, (c) λ_{tv}, and (d) β) in the proposed method for HSI CS on the PaviaUA and Toy data sets.

noise remains. LRMR is the second-best and hardly handles heavy Gaussian cases. 2DTV and E-3DTV spend similar computational time because the update processing of 2DTV is simple, and E-3DTV converges quickly. HSSTV uses two local differences and has a slow convergence, which leads to an expensive computation cost. In contrast, although l_0-l_1HTV contains two TV norms, l_0hTV and SSTV, the running times of l_0-l_1HTV and SSTV are similar since the pure SSTV model converges slowly. For HSI CS, ASSTV, SparCS, and HSSTV are more computationally expensive. Due to its fast convergence speed, E-3DTV is the fastest one among all the CS algorithms. Our proposed method is slightly slower than SSTV. However, l_0-l_1HTV obtains the best performances in the HSI restoration experiments.

VII. CONCLUSION

In this article, we have proposed a novel l_0-l_1HTV regularization for HSI denoising and CS restoration. l_0 HS TV restrains the spatial smoothness from all spectral bands, and SSTV exploits local structure information along with spatial and spectral directions. The hybrid TV is designed as a globally and locally integrated regularizer for HSI restorations. Experimental results on HSI mixed noise removal and CS recovery demonstrate the superior performances of our proposed methods.

For further work, two extensions of our regularization technique can be explored: 1) l_0-l_1HTV can be applied to other HSI processing tasks, such as classification and 2) in this study, the simple use of l_0-l_1HTV has been focused. In fact, this kind of HTV can be injected into LR matrix or tensor models.

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